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C9555 A Survival Guide to Simulations

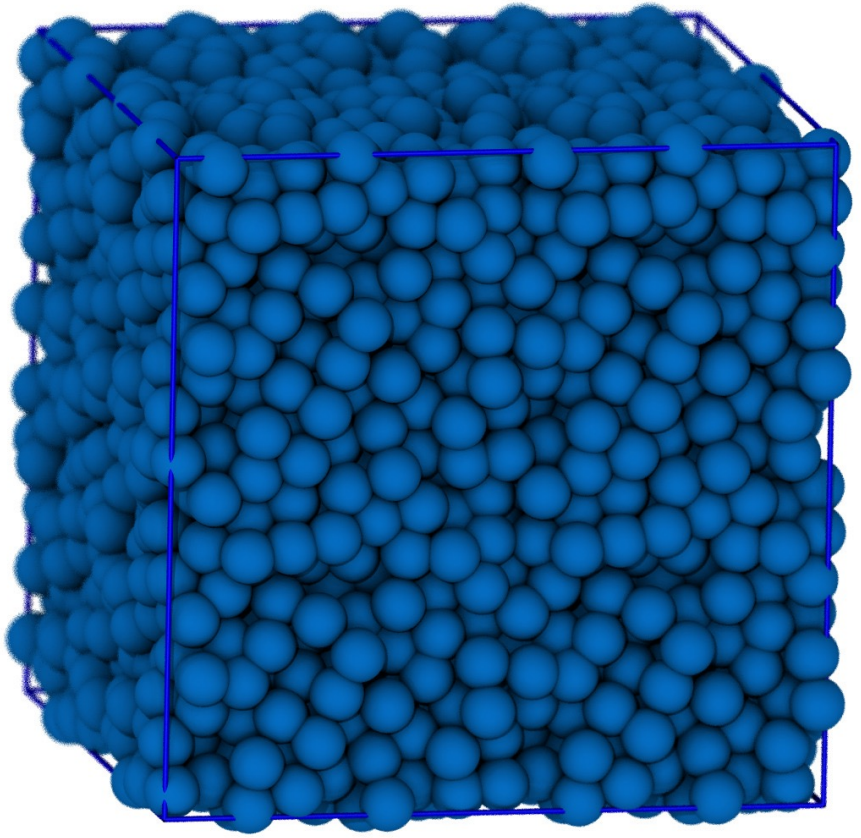
Free Energy Calculations, Part 2

Ivo Kabelka

Last Week's Task 1

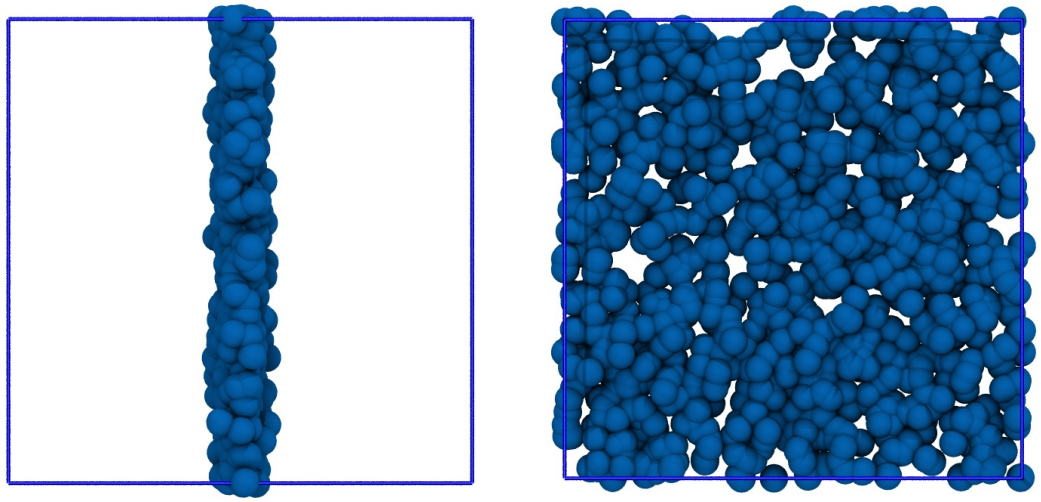
Water box

- Restrain a bead
  - Within a plane
  - Within a slab
  - Along an axis
  - Within a cylinder



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Restrain a Bead Within a Plane



```
c: CENTER ATOMS=1-3200 NOPBC

d: DISTANCE ATOMS=c,1 SCALED_COMPONENTS

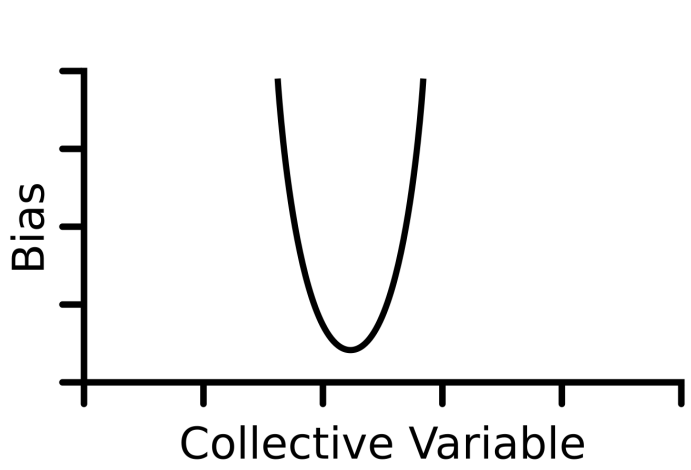
res: RESTRAINT ARG=d.a AT=0.0 KAPPA=8000

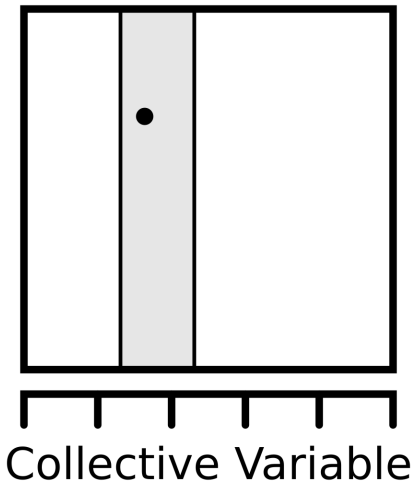
PRINT ARG=d.a,res.bias FILE=COLS
```

dat (plumed)

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Harmonic Restraint in PLUMED



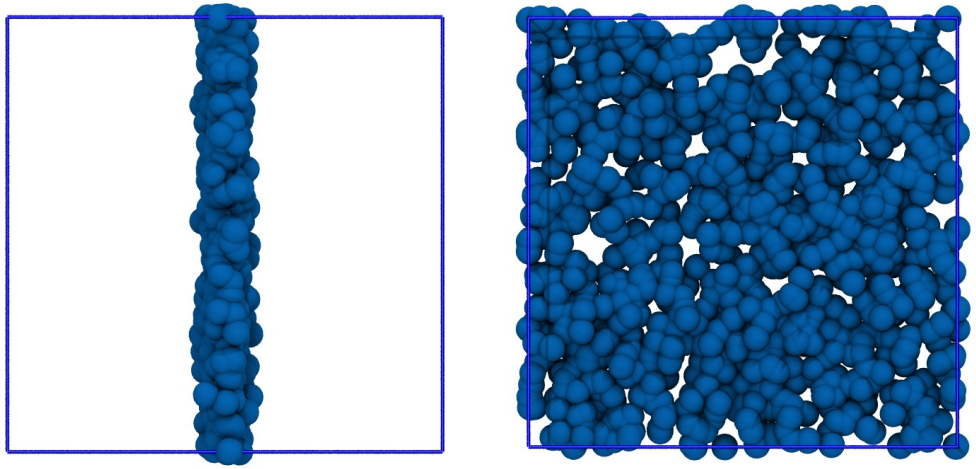



```
res: RESTRAINT ARG=dist.z AT=0.0 KAPPA=8000
```
dat (plumed)

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Restrain a Bead Within a Plane

Centering the first water bead within the box

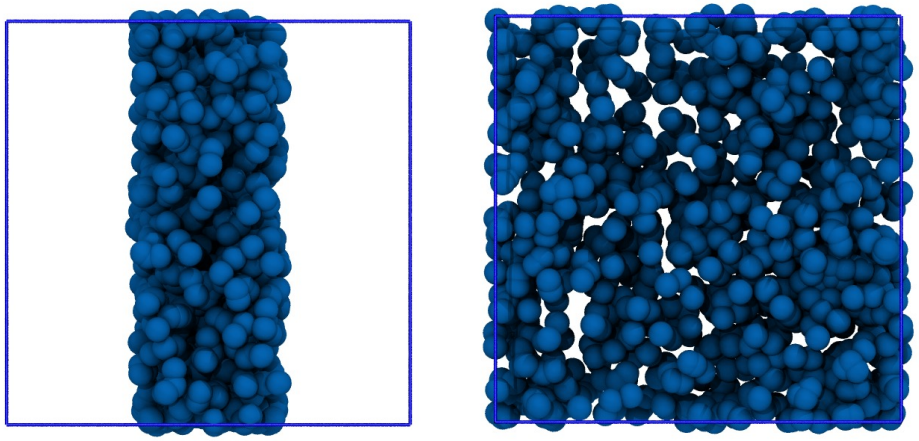


```
$ gmx trjconv -s conf.gro \
              -f conf.gro \
              -n index.ndx \
              -o conf_new.gro \
              -pbc atom \
              -center
```

bash

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Restrain a Bead Within a Slab



```
c: CENTER ATOMS=1-3200 NOPBC

d: DISTANCE ATOMS=c,1 SCALED_COMPONENTS

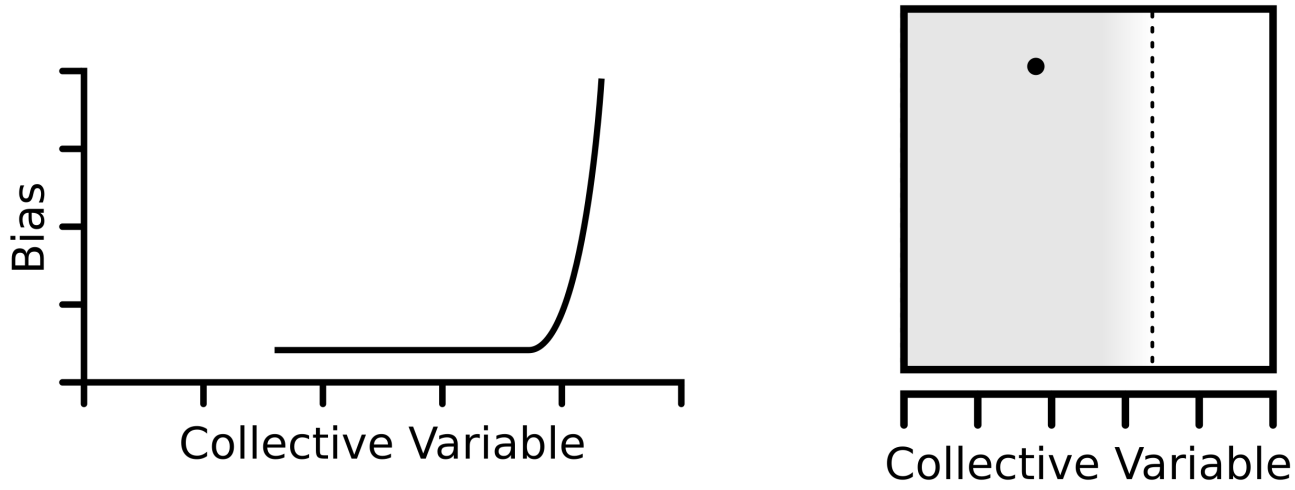
lwall: LOWER_WALLS ARG=d.a AT=-0.15 KAPPA=8000 EXP=2 EPS=1 OFFSET=0
uwall: UPPER_WALLS ARG=d.a AT=0.15  KAPPA=8000 EXP=2 EPS=1 OFFSET=0

PRINT ARG=d.a,lwall.bias,uwall.bias FILE=COLS
```

dat (plumed)

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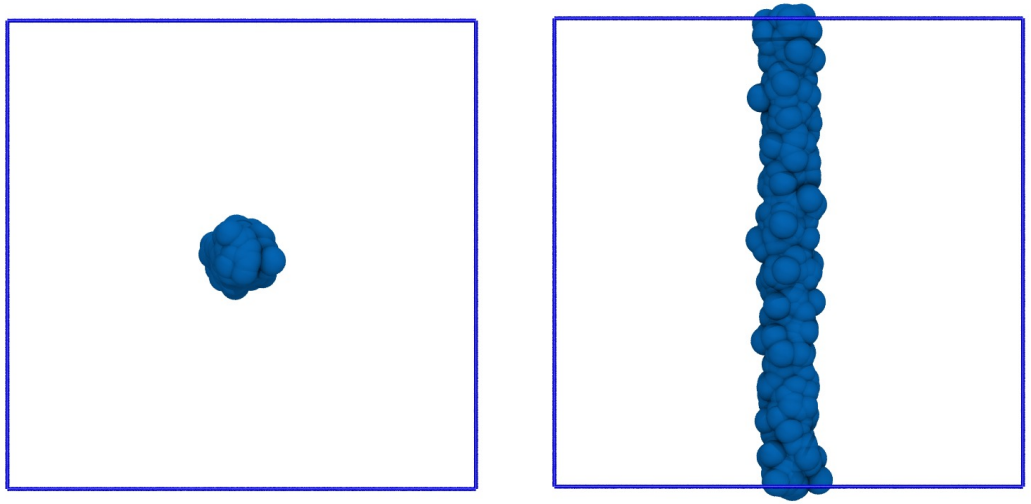
Flat-bottom Restraints in PLUMED




```
res: UPPER_WALLS ARG=dist.z AT=0.0 KAPPA=8000
```
dat (plumed)

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Restrain a Bead Along an Axis



```
c: CENTER ATOMS=1-3200 NOPBC

d: DISTANCE ATOMS=c,1 SCALED_COMPONENTS

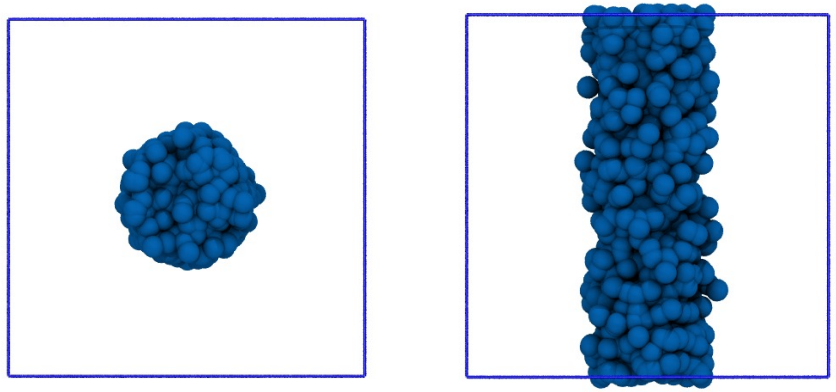
res: RESTRAINT ARG=d.a,d.b AT=0.0,0.0 KAPPA=8000,8000

PRINT ARG=d.a,d.b,res.bias FILE=COLS
```

dat (plumed)

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Restrain a Bead Within a Cylinder



```
c: CENTER ATOMS=1-3200 NOPBC

d: DISTANCE ATOMS=c,1 SCALED_COMPONENTS

dist: CUSTOM ARG=d.a,d.b FUNC=sqrt(x^2+y^2) PERIODIC=NO

uwall: UPPER_WALLS ARG=dist AT=0.15 KAPPA=8000 EXP=2 EPS=1 OFFSET=0

PRINT ARG=dist,uwall.bias FILE=COLS
```

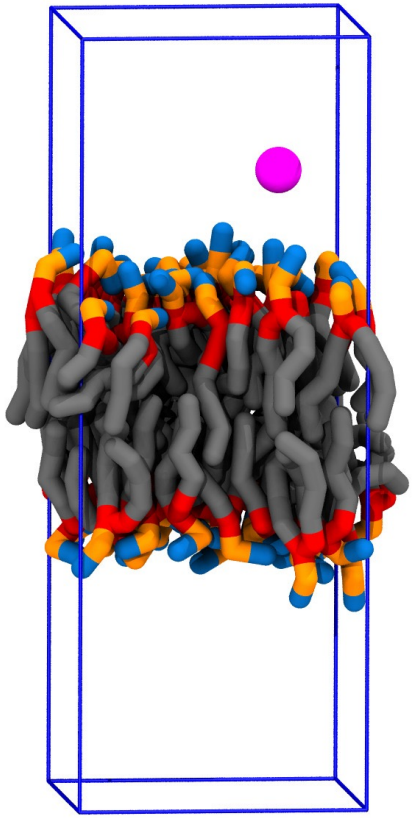
dat (plumed)

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Last Week's Task 2

Adsorption of calcium ion on a lipid bilayer

- Perform an unbiased MD simulation
- Calculate the free energy of adsorption



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Calculate Free Energy

```
popc: GROUP NDX_FILE=index.ndx NDX_GROUP=POPC UNIQUE
ca: GROUP NDX_FILE=index.ndx NDX_GROUP=CA UNIQUE

center: CENTER ATOMS=popc NOPBC

d: DISTANCE ATOMS=center,ca SCALED_COMPONENTS
dist: CUSTOM ARG=d.c FUNC=x PERIODIC=-0.5,0.5

h: HISTOGRAM ARG=dist GRID_MIN=-0.5 GRID_MAX=0.5 GRID_BIN=100 BANDWIDTH=0.01

fes: CONVERT_TO_FES GRID=h TEMP=310
DUMPGRID GRID=fes FILE=fes.dat

PRINT STRIDE=1   ARG=dist,d.c FILE=DIST
FLUSH STRIDE=100
```

dat (plumed)

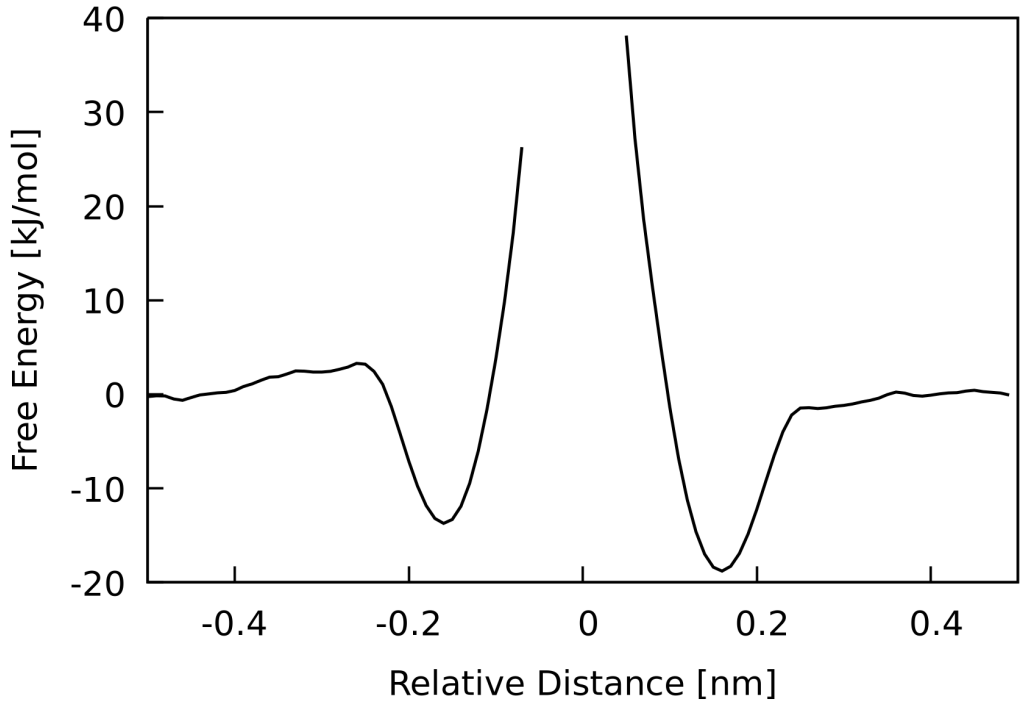
```
$ plumed driver --plumed plumed.dat --mf_xtc prod.xtc
```

bash

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Calculate Free Energy

Simulation length: 15 μs



gnuplot

```
set terminal pngcairo

unset key
set border lw 4

set xlabel "Relative Distance [nm]"
set ylabel "Free Energy [kJ/mol]"

set xrange [-0.5:0.5]
set yrange [-20:40]

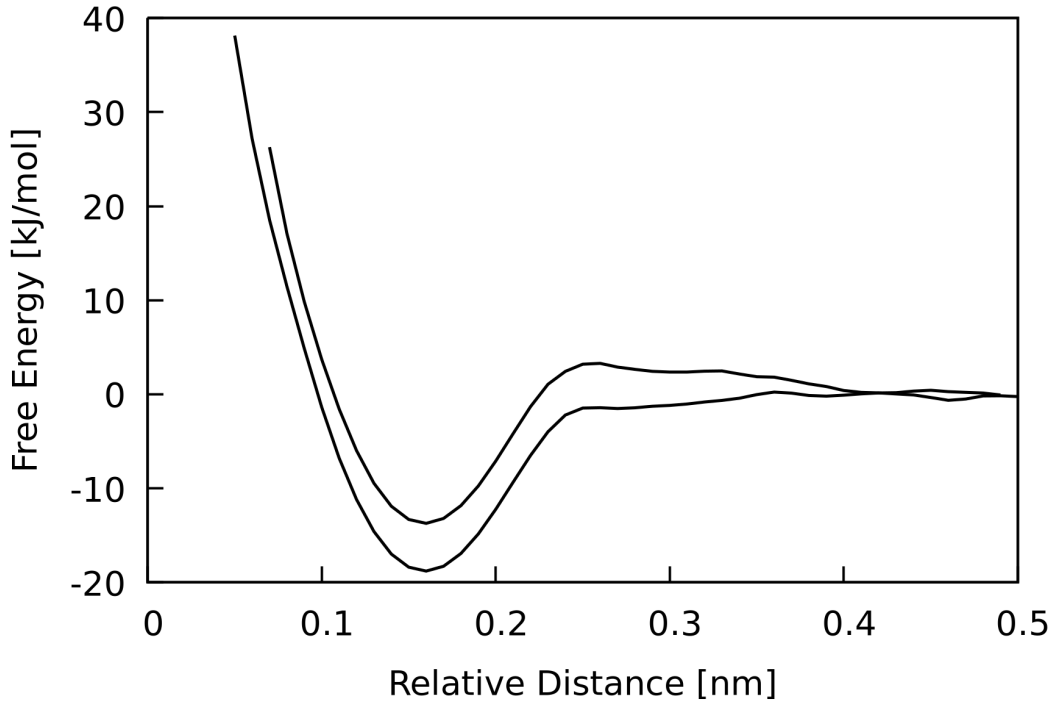
set xtics nomirror
set ytics nomirror

set output "name.png"
plot 'fes.dat' w l lc rgb "black"
```


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Calculate Free Energy

Simulation length: 15 μs



gnuplot

```
set terminal pngcairo

unset key
set border lw 4

set xlabel "Relative Distance [nm]"
set ylabel "Free Energy [kJ/mol]"

set xrange [-0.5:0.5]
set yrange [-20:40]

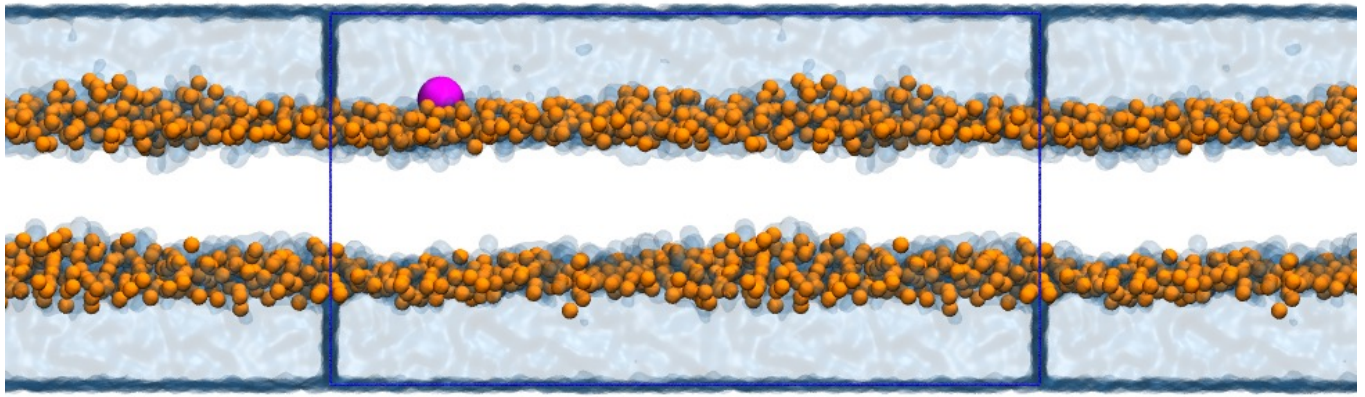
set xtics nomirror
set ytics nomirror

set output "name.png"
plot 'fes.dat' u ($1<0 ? -$1 : $1):2
```


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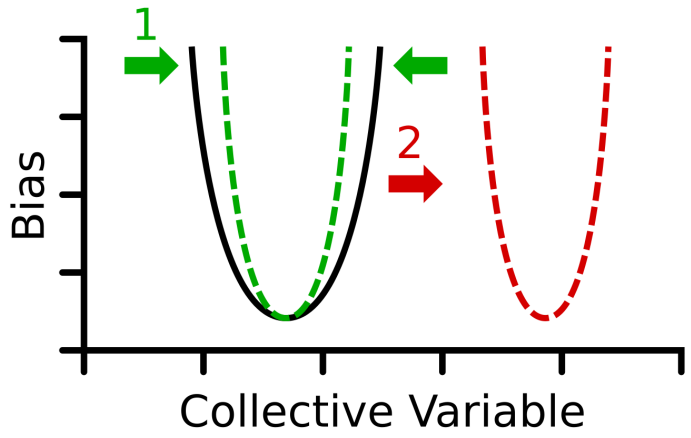
Last Week's Task 3

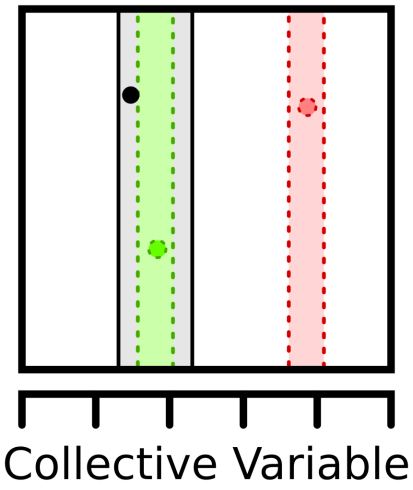
- Perform a pulling simulation using GROMACS and PLUMED
  - Pull an ion across the membrane (in Z-axis)
  - Use pull code (GROMACS) and MOVINGRESTRAINT (PLUMED)



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Moving Restraints in PLUMED






```
MOVINGRESTRAINT ...
    LABEL=res
    ARG=dist.z
    STEP0=0          AT0=-2.5   KAPPA0=2500
    STEP1=10000      AT1=-2.5   KAPPA1=8000
    STEP2=5000000    AT2=2.5    KAPPA2=8000
...
```

dat (plumed)

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Pulling Simulation

```
popc: GROUP NDX_FILE=index.ndx NDX_GROUP=POPC UNIQUE
ca: GROUP NDX_FILE=index.ndx NDX_GROUP=CA UNIQUE

center: CENTER ATOMS=popc NOPBC

dist: DISTANCE ATOMS=center,ca COMPONENTS

MOVINGRESTRAINT ...
    LABEL=res
    ARG=dist.z
    STEP0=0         AT0=-2.5  KAPPA0=500
    STEP1=10000     AT1=-2.5  KAPPA1=8000
    STEP2=5000000   AT2=2.5   KAPPA2=8000
...

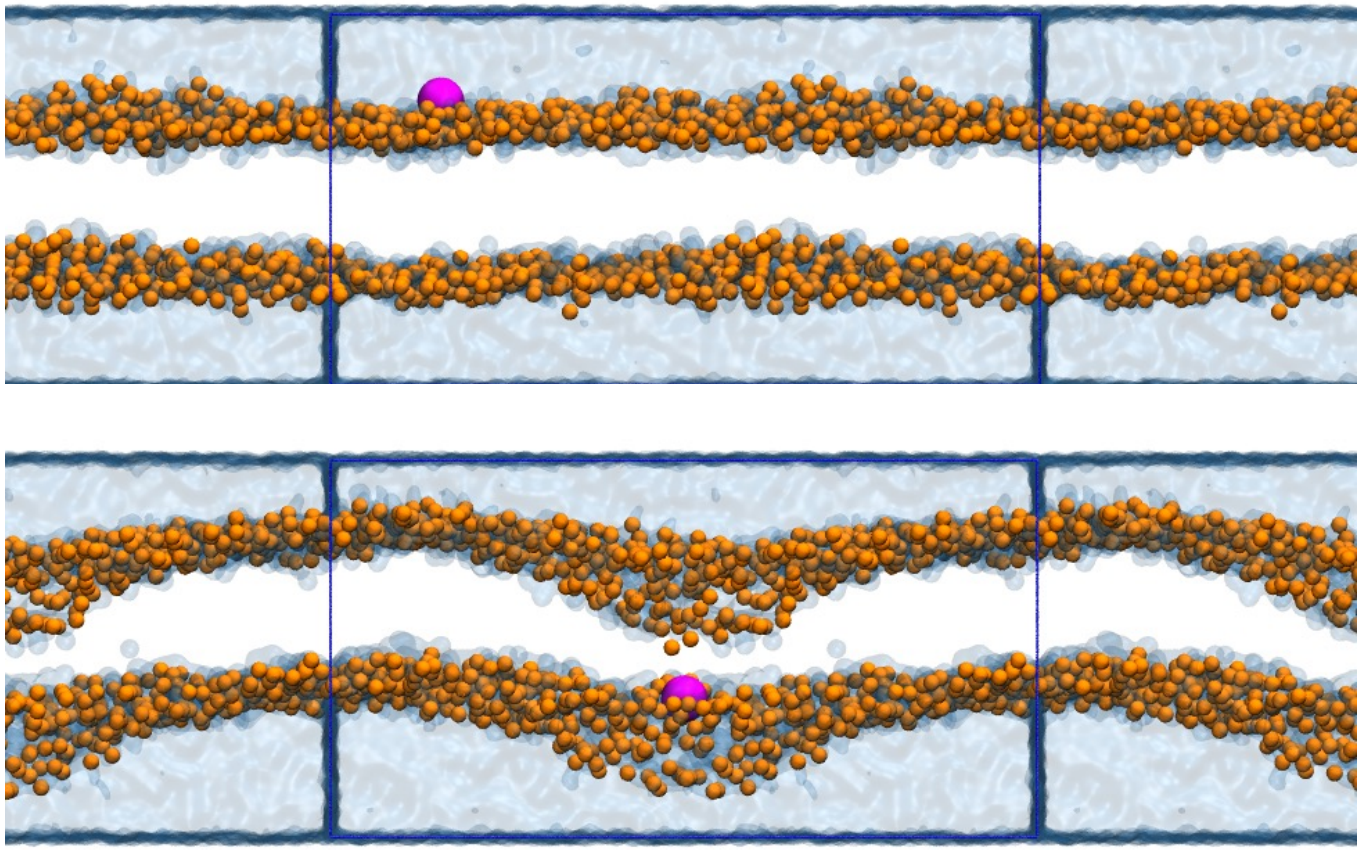
PRINT STRIDE=5000 ARG=rest.bias,dist.z FILE=DIST
FLUSH STRIDE=1000
```

dat (plumed)

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Pulling Simulation

The collective variable does not describe the depth of ion insertion



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Pulling Simulation

Calculating a local membrane center of mass in GROMACS

```
pull                    = yes
pull-nstxout            = 1

pull_ngroups            = 2
pull_ncoords            = 1
pull_group1_name        = POPC
pull_group2_name        = CA

pull-cylinder-r         = 1.0
pull_coord1_type        = umbrella
pull_coord1_geometry    = cylinder
pull_coord1_groups      = 1 2
pull_coord1_vec         = 0 0 1
pull_coord1_dim         = N N Y
```

mdp (gromacs)

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Pulling Simulation

Calculating a local membrane center of mass in GROMACS

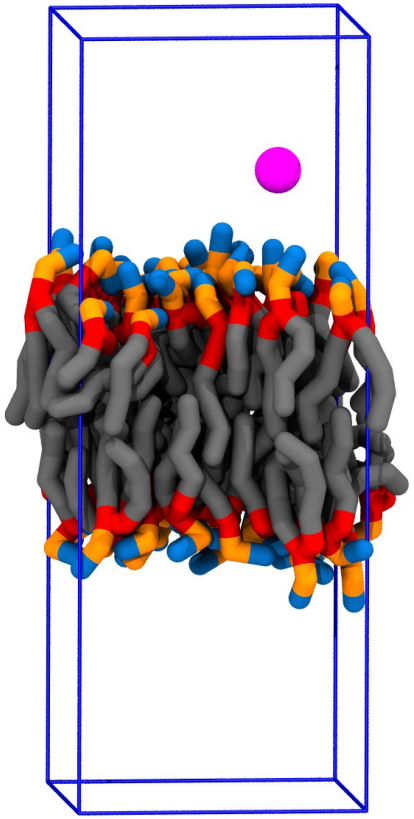
– A tpr file is needed for the calculation

```bash
$ gmx grompp -f rerun.mdp \
             -n index.ndx \
             -p topol.top \
             -c equi.gro \
             -o rerun.tpr

$ gmx mdrun -deffnm rerun \
            -rerun pull.xtc
```

Pulling Simulation

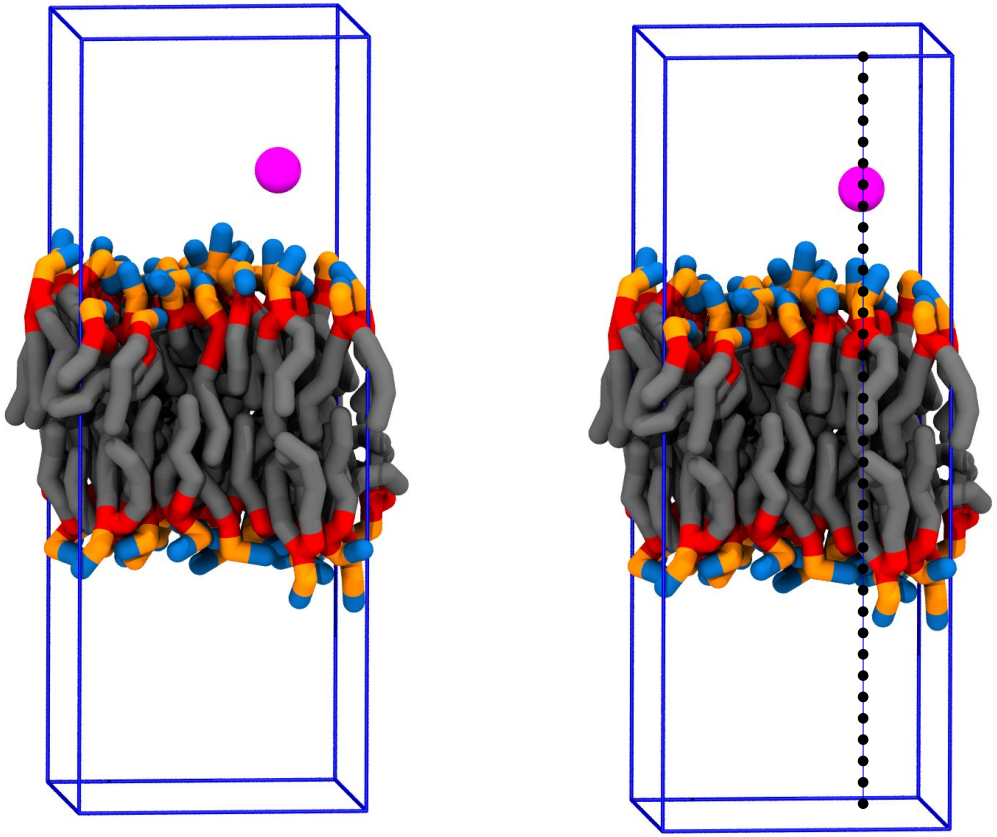
Calculating a local membrane center of mass in GROMACS



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Pulling Simulation

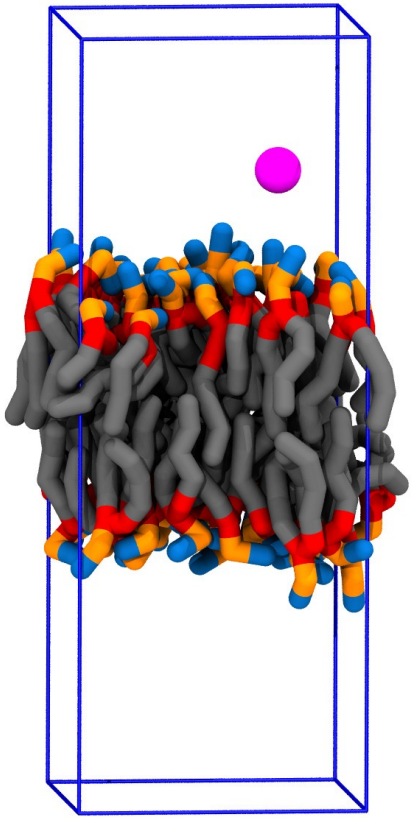
Draw an axis along **`pull_coord1_vec`** going through **`pull_group2_name`**

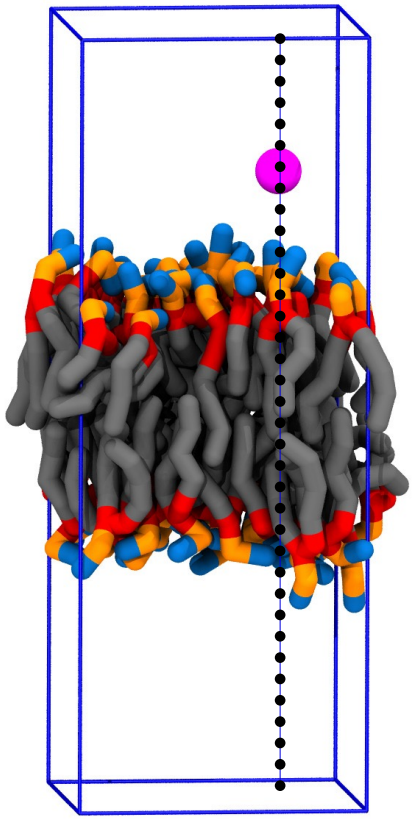


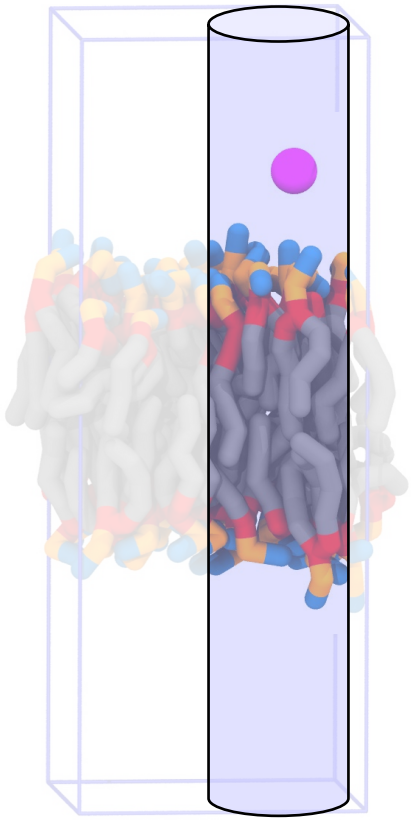
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Pulling Simulation

Select atoms of **`pull_group1_name`** within **`pull-cylinder-r`**



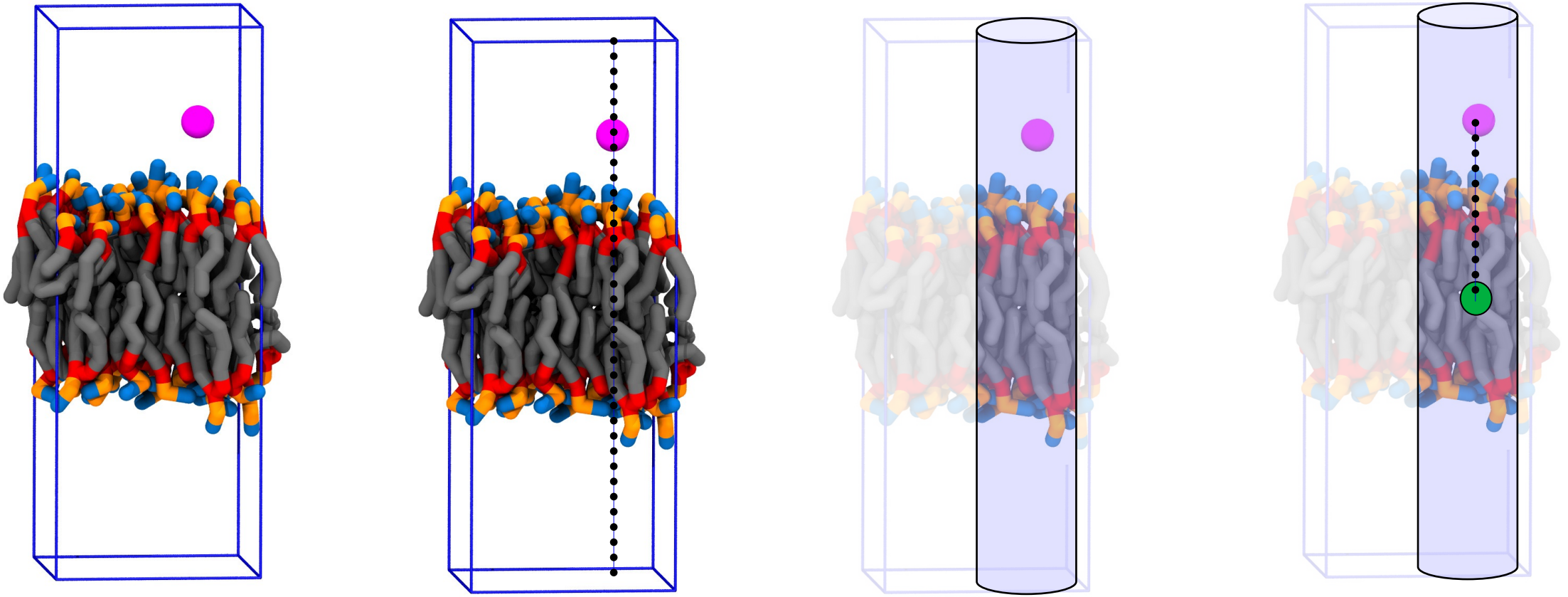




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Pulling Simulation

Calculate the distance between the centers of mass of both selections



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Pulling Simulation

Calculating a local membrane center of mass in PLUMED

```
popc: GROUP NDX_FILE=index.ndx NDX_GROUP=POPC UNIQUE
ca: GROUP NDX_FILE=index.ndx NDX_GROUP=CA UNIQUE

ZDISTANCES GROUPA=popc GROUPB=ca LABEL=dist NOPBC LOWMEM

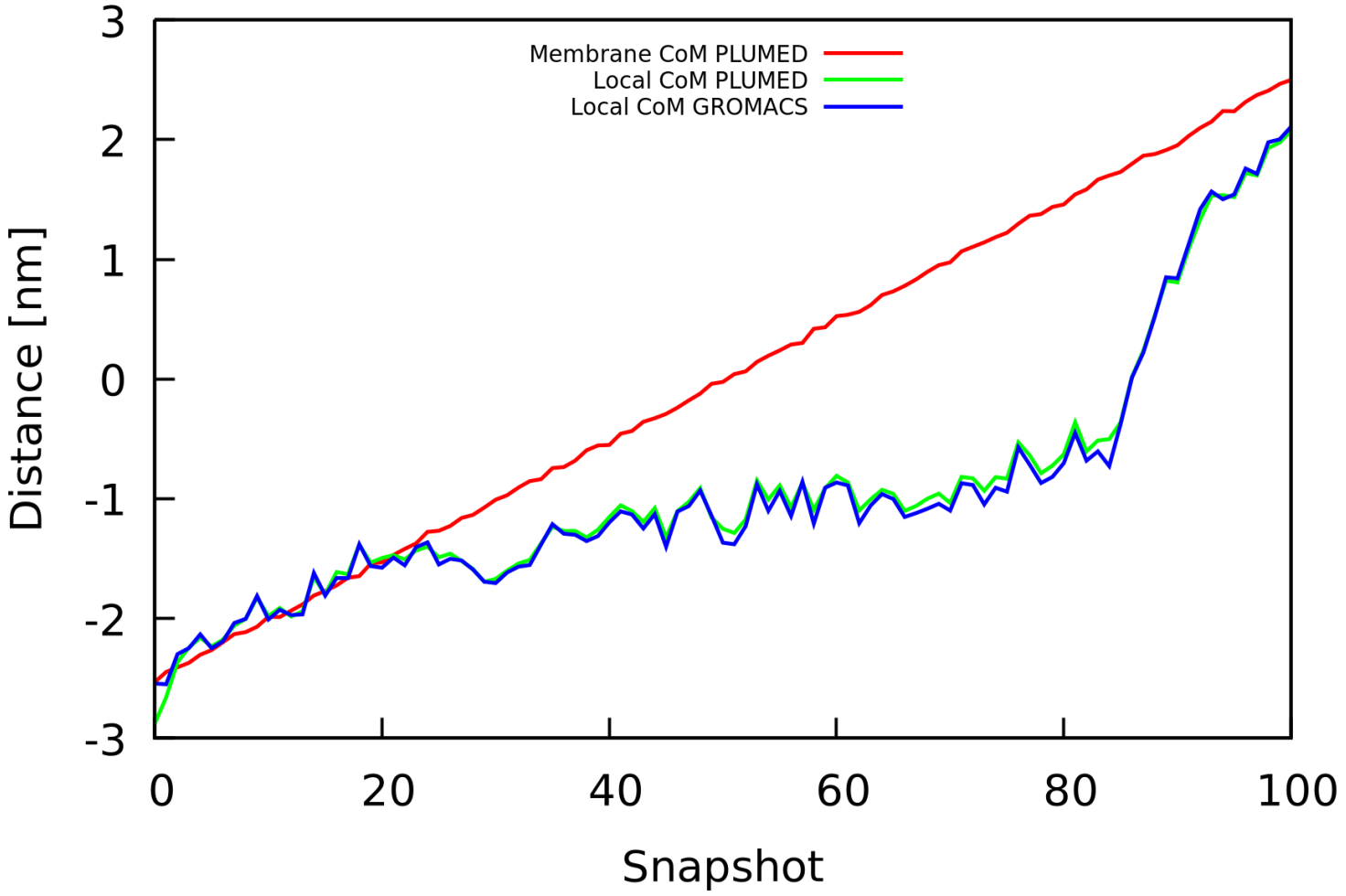
INCYLINDER ATOM=ca DATA=dist DIRECTION=Z RADIUS={RATIONAL R_0=0.4 NN=4 …
MM=12 D_MAX=1.0} LOWER=0.0 UPPER=0.0 SIGMA=0.1 LABEL=distc MEAN LOWMEM

PRINT STRIDE=1 ARG=distc.mean FILE=DIST_2
FLUSH STRIDE=100
```

dat (plumed)

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Pulling Simulation



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Umbrella Sampling

```bash
$ mkdir -p snapshots

$ gmx trjconv -s pull.tpr \
              -f pull.trr \
              -o snapshots/snap..gro \
              -sep \
              -split 0 <<< 0
```

bash

```
pull-nstxout                = 100
pull-nstfout                = 100

pull_coord1_type            = umbrella
pull_coord1_rate            = 0.0
pull_coord1_k               = 8000
pull_coord1_start           = no
pull_coord1_init            = 0.0
```

mdp (gromacs)

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WHAM

```
$ ls -1 ??/prod.tpr > wham_tpr.dat
$ ls -1 ??/prod_pullx.xvg > wham_xvg.dat

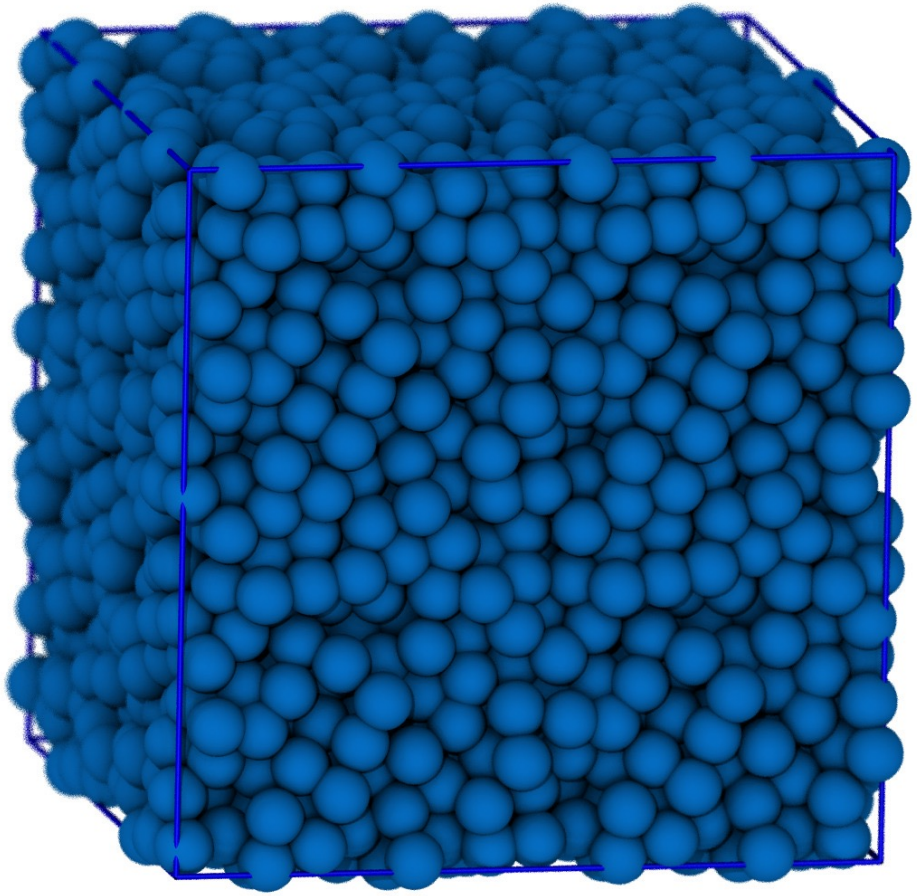
$ gmx wham -it wham_tpr.dat \
           -ix wham_xvg.dat \
           -o fes.xvg \
           -hist hist.xvg \
           -unit kJ \
           -min -2.5 \
           -max 2.5 \
           -temp 310 \
           -b 10000 # -e 1000000
```

bash

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The Toy System

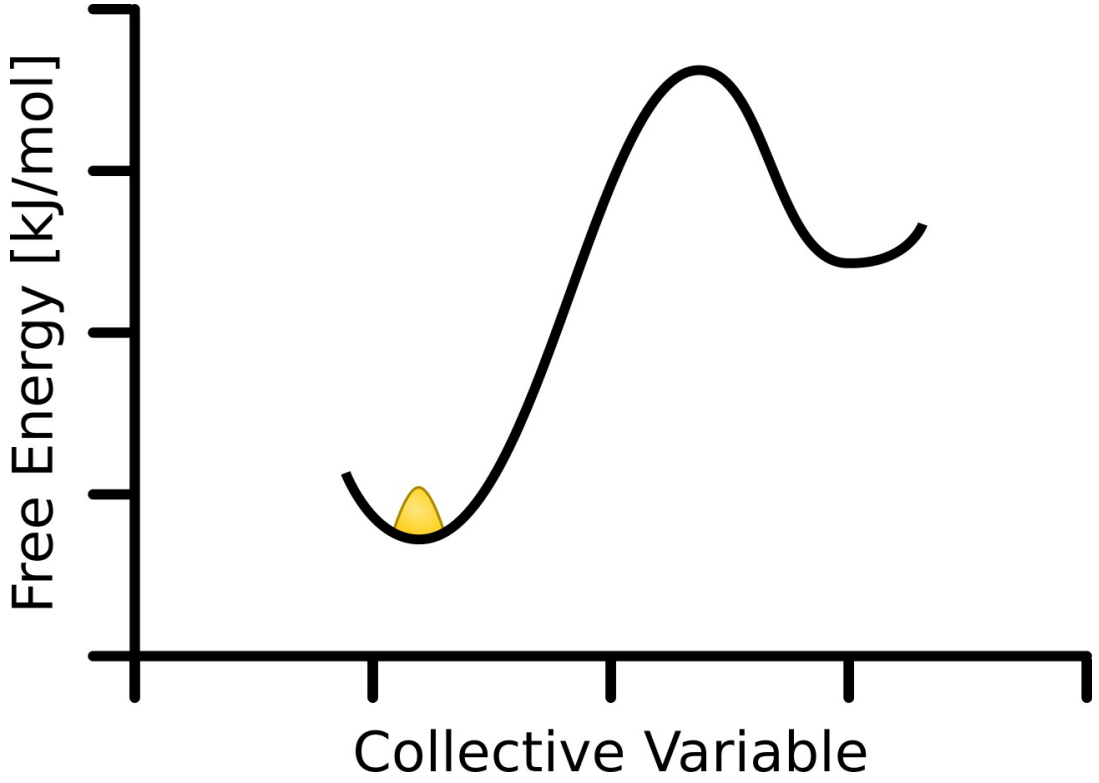
Water box



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Metadynamics

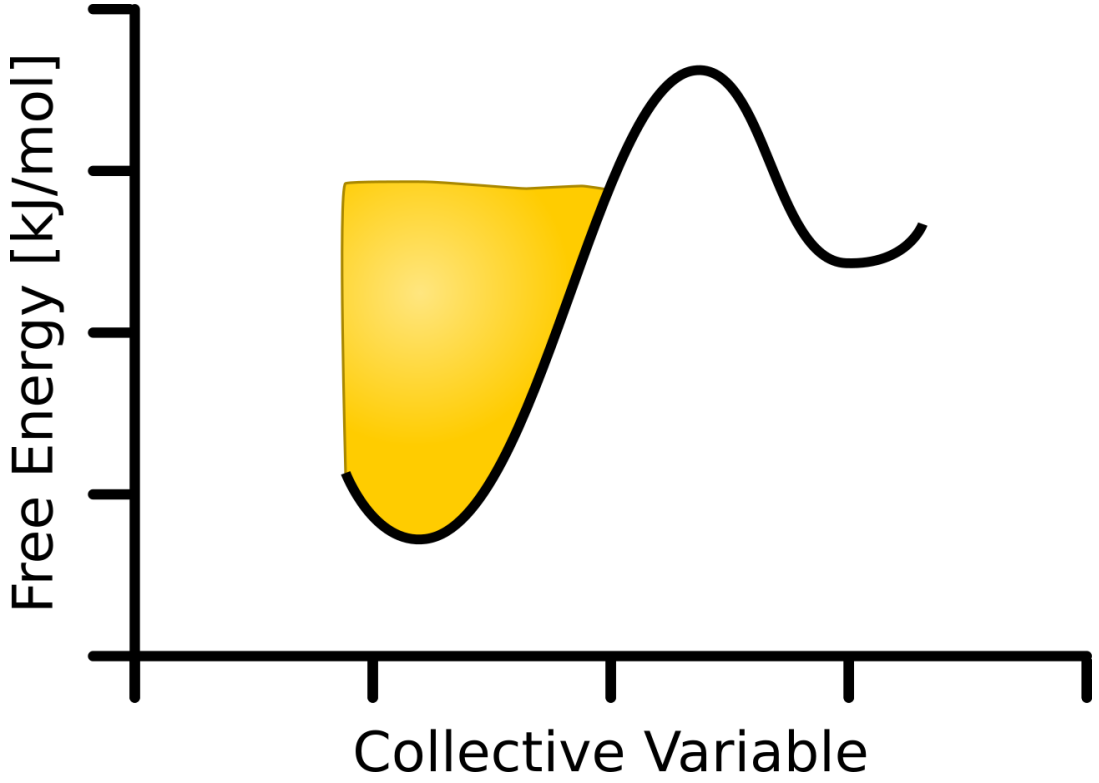
Filling the free energy wells with computational sand



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Metadynamics

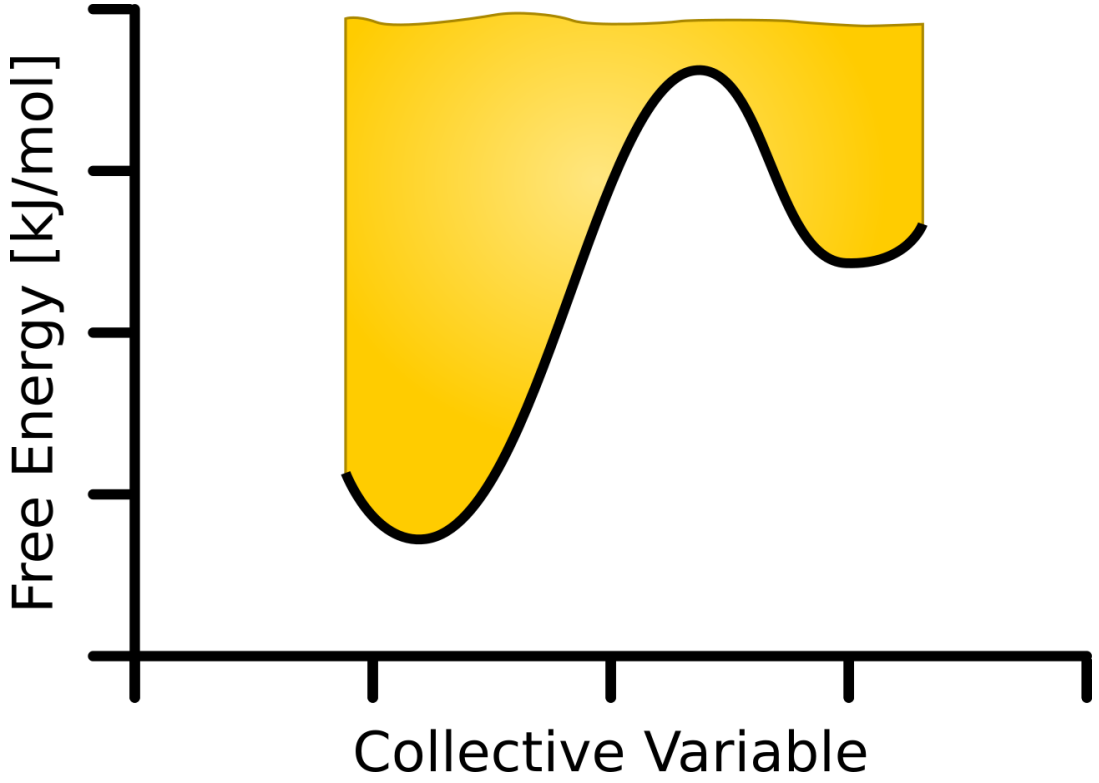
A biasing potential is added at a selected rate to “fill” already visited regions of the CV …



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Metadynamics

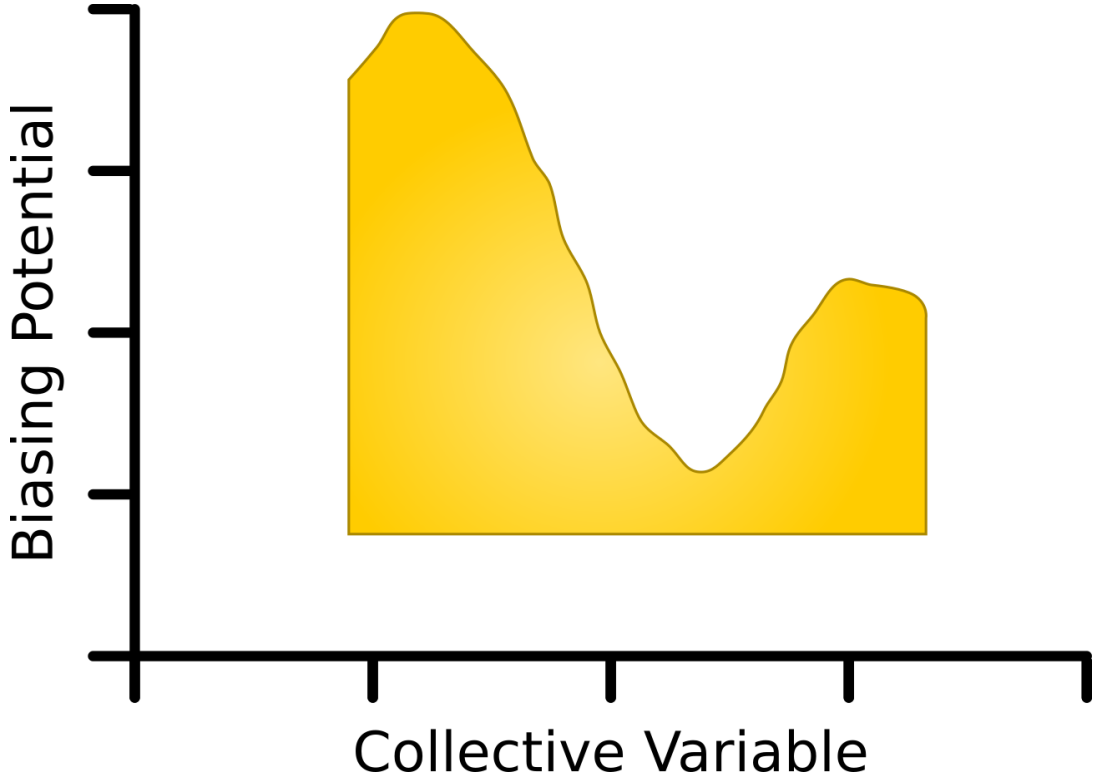
… to effectively lower free energy barriers, enabling the exploration of new regions in the CV



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Metadynamics

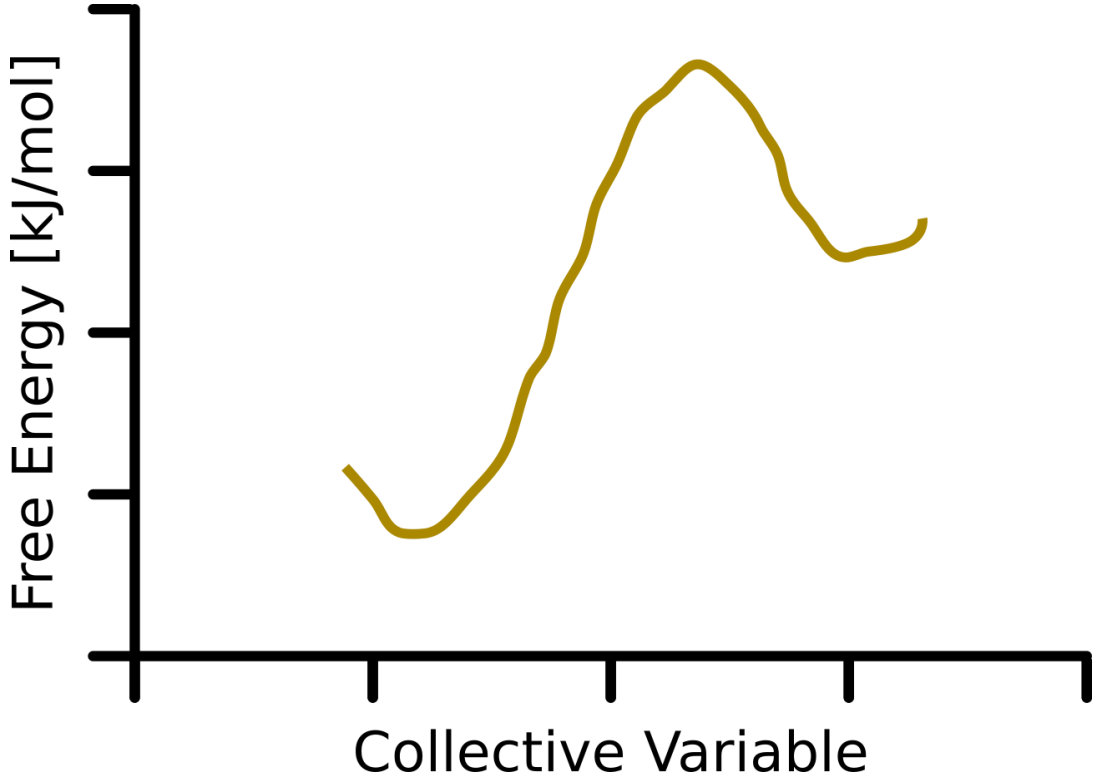
The accumulated biasing potential is summed …



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Metadynamics

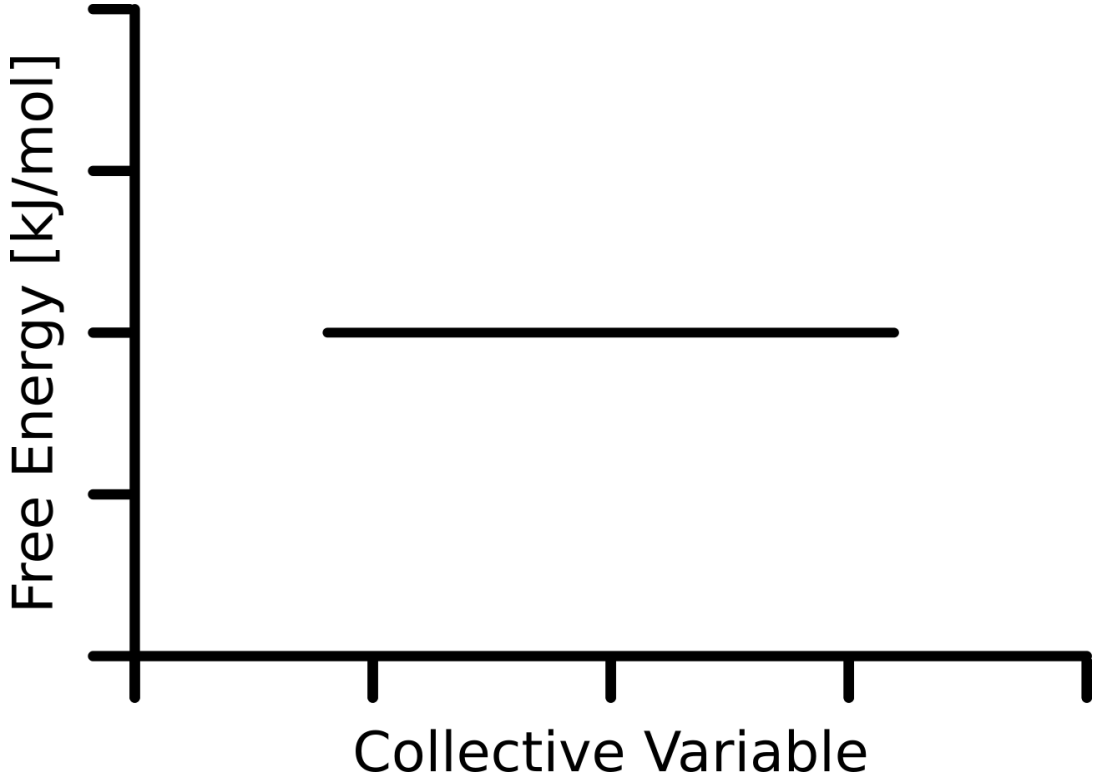
… and its negative value represents the free energy



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Metadynamics

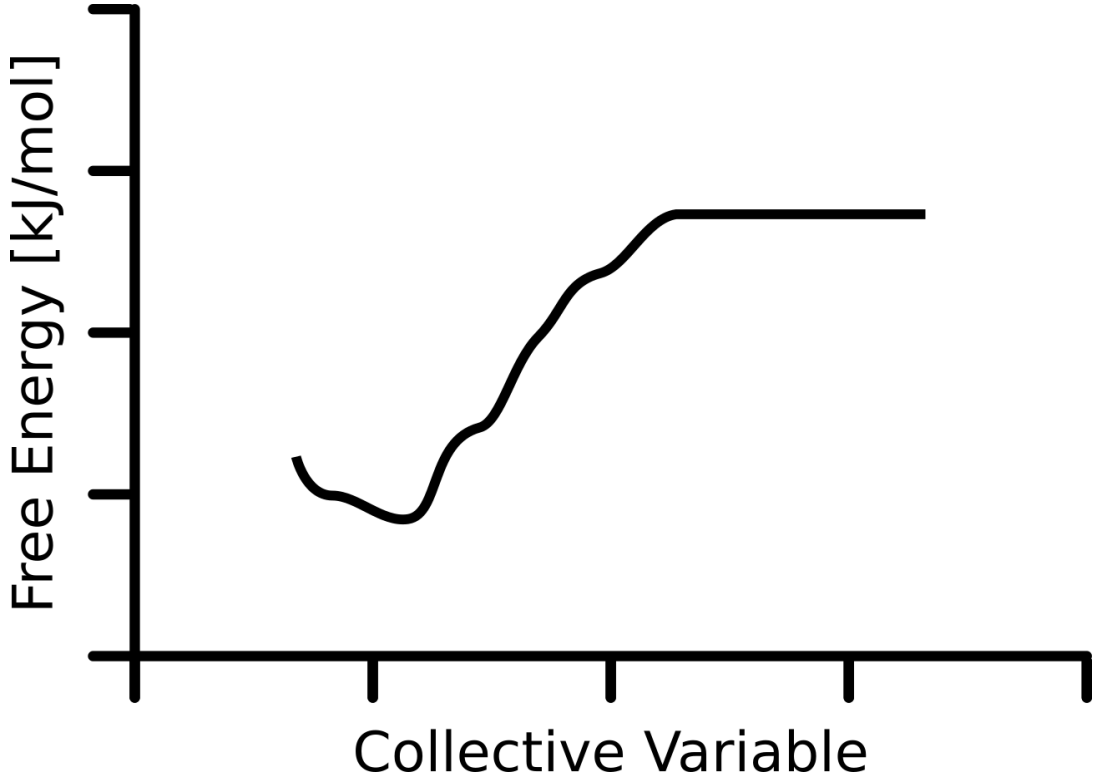
Evolution of the calculated free energy profile



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Metadynamics

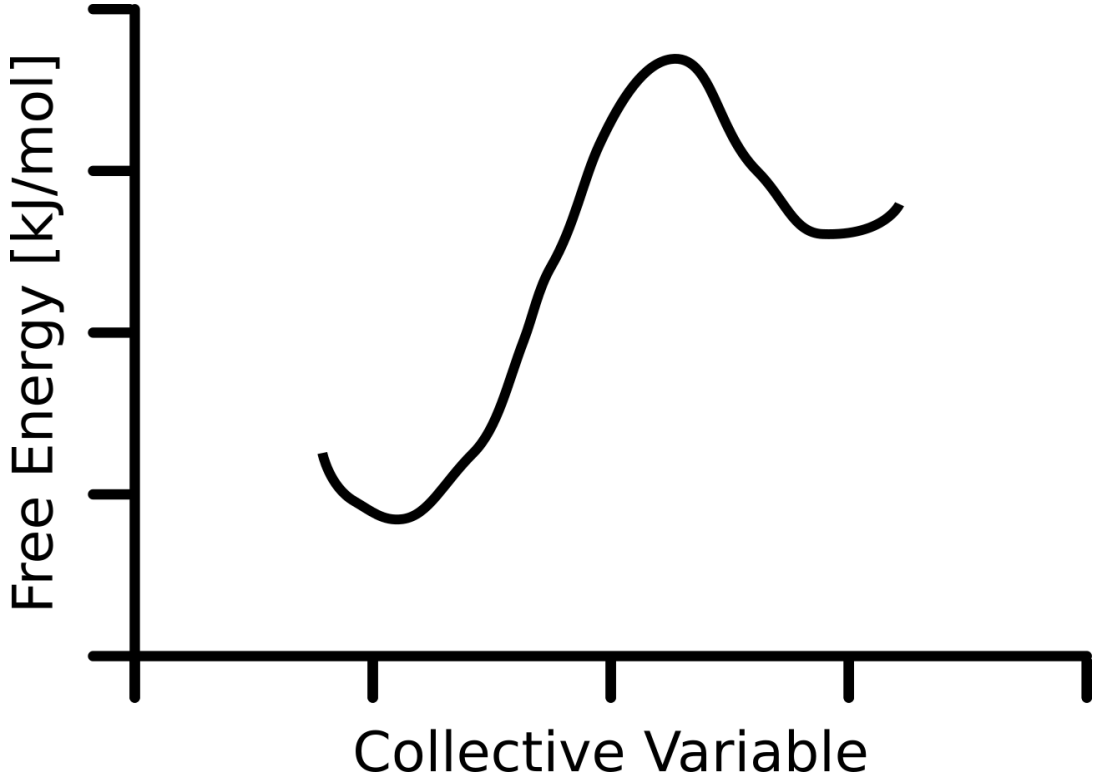
Evolution of the calculated free energy profile



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Metadynamics

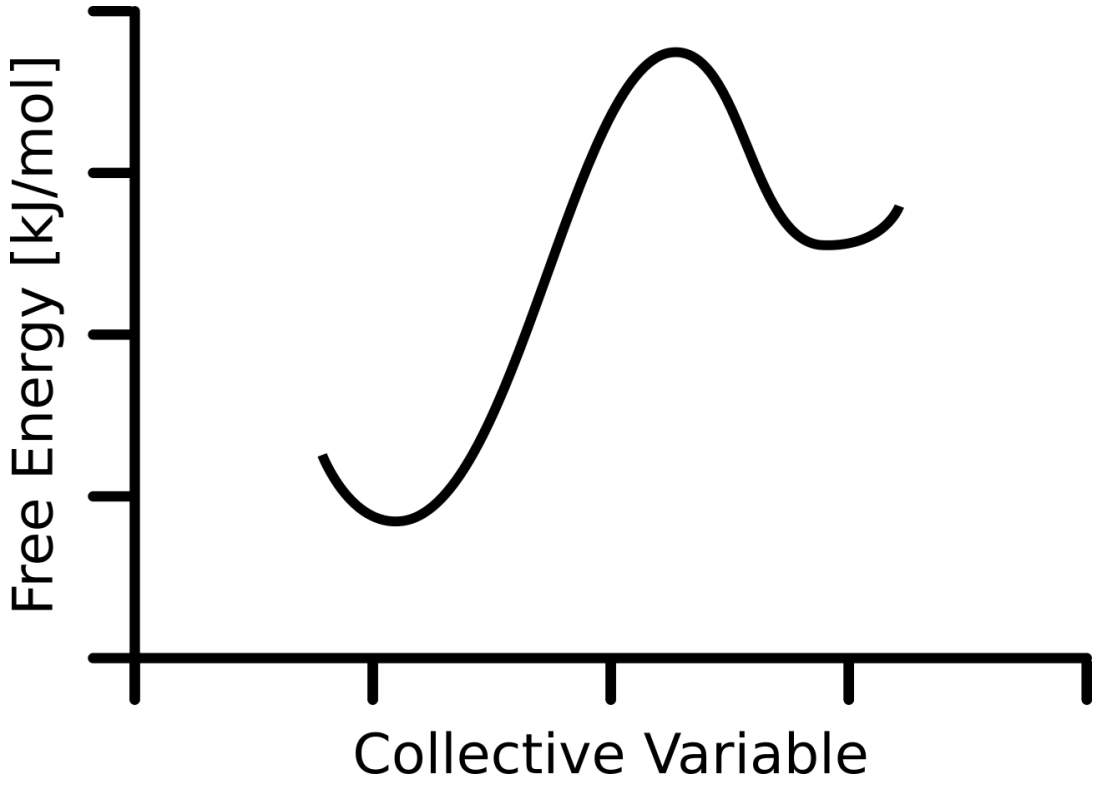
Evolution of the calculated free energy profile



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Metadynamics

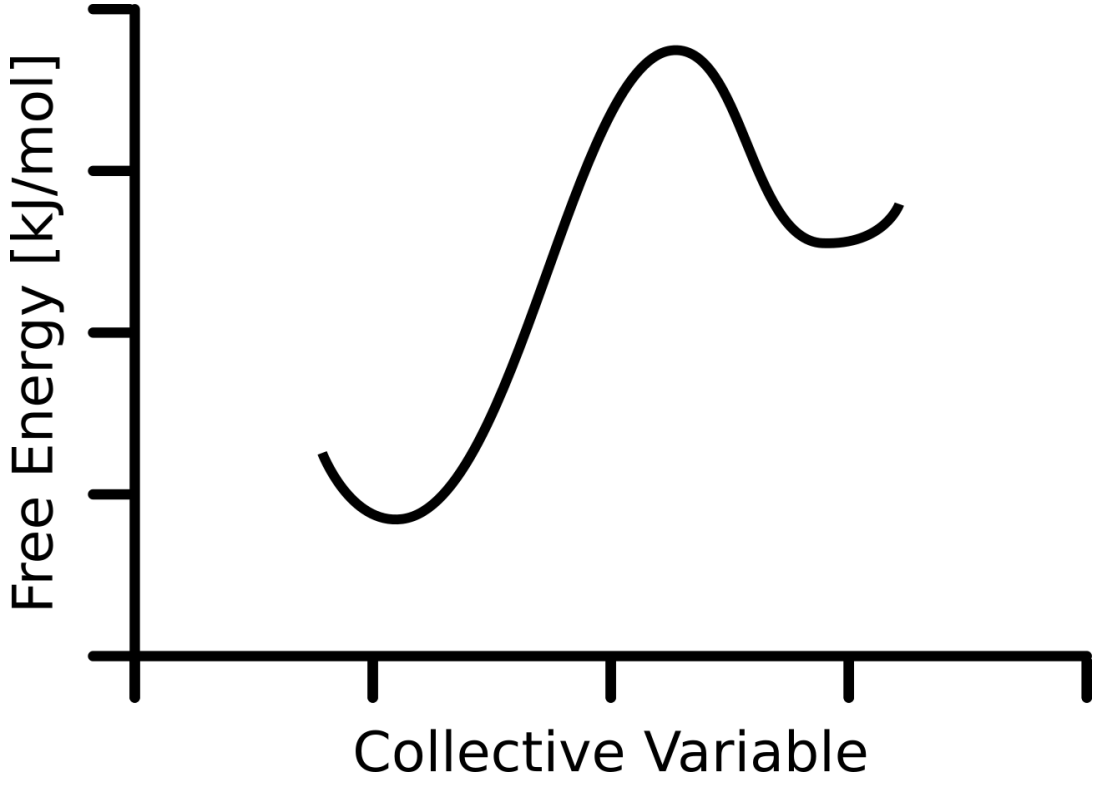
Evolution of the calculated free energy profile



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Metadynamics

Using `sum_hills` to obtain the free energy profile

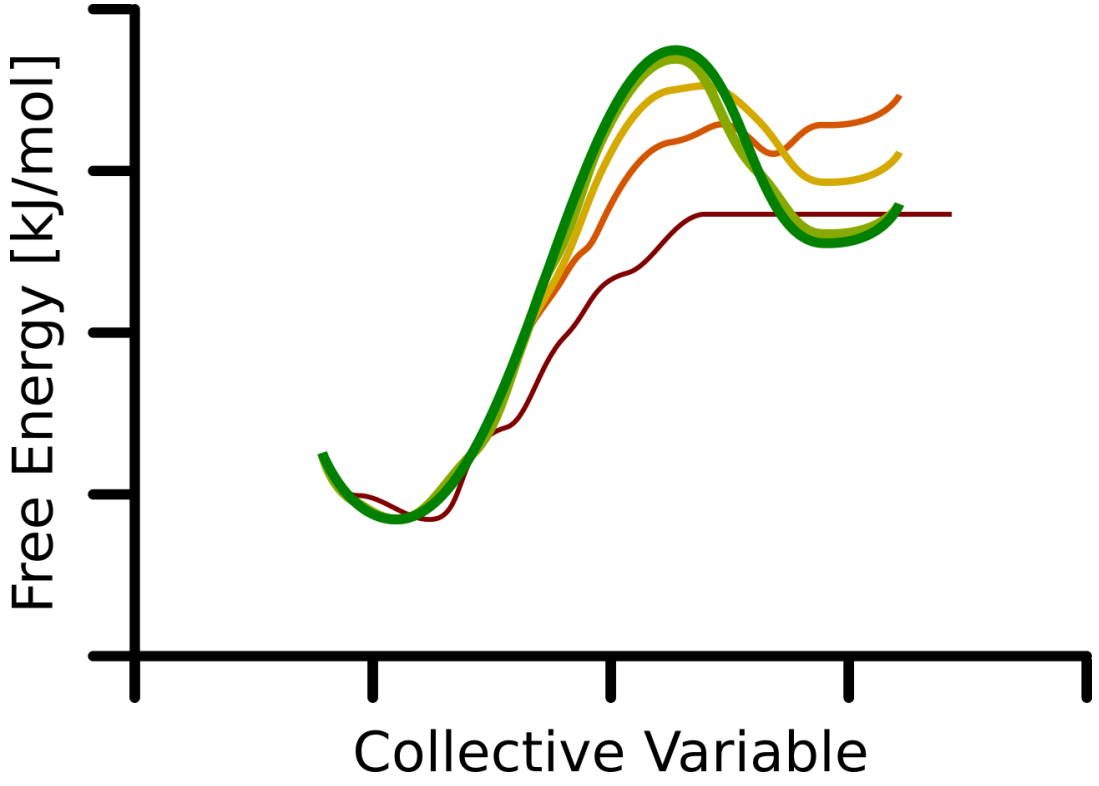



```
$ plumed sum_hills --hills HILL --mintozero
```
bash

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Metadynamics

Using **`sum_hills`** to obtain the time evolution of the free energy profile




```
$ plumed sum_hills --hills HILL --mintozero --stride 1000
```
bash

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Metadynamics

Several parameters must be set

```
METAD ...
    LABEL=mtd
    ARG=dist

    SIGMA=0.05
    HEIGHT=1
    PACE=500

    GRID_MIN=-0.5
    GRID_MAX=0.5
    GRID_SPACING=0.01

    FILE=HILL
...
```

dat (plumed)

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Metadynamics

For computational efficiency, the Gaussians can be deposited on a grid

```
METAD ...
    LABEL=mtd
    ARG=dist

    SIGMA=0.05
    HEIGHT=1
    PACE=500

    GRID_MIN=-0.5
    GRID_MAX=0.5
    GRID_SPACING=0.01

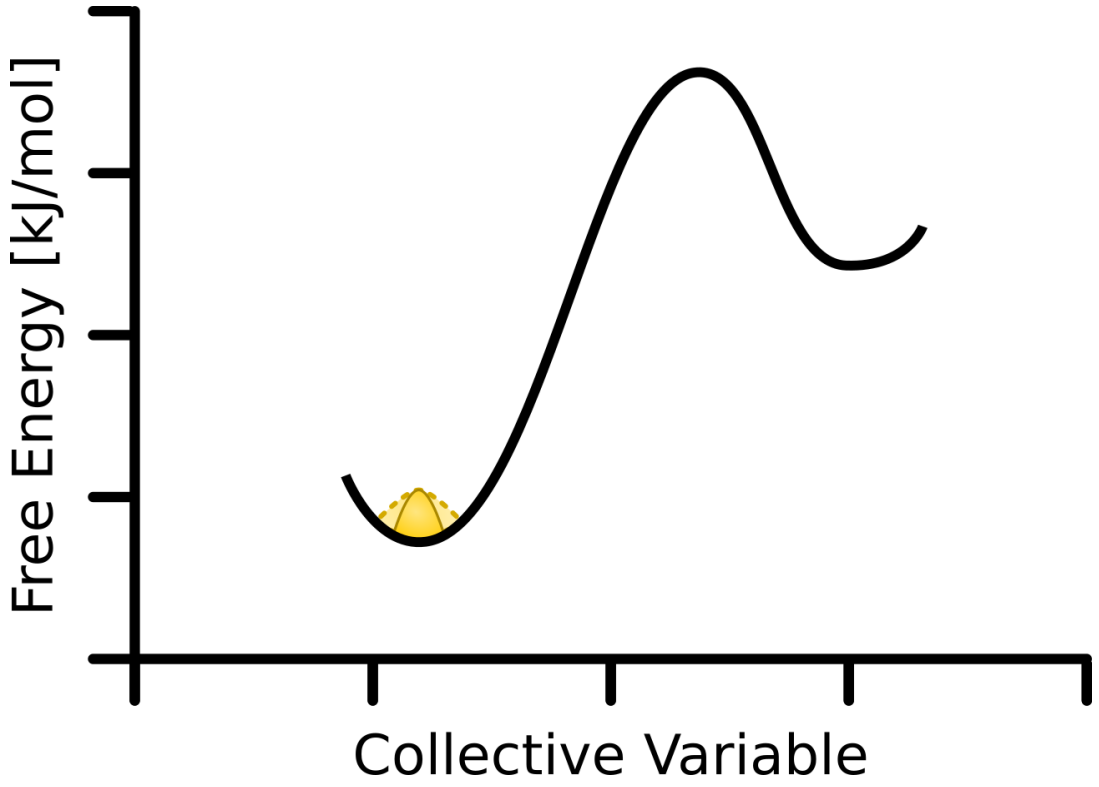
    FILE=HILL
...
```

dat (plumed)

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Metadynamics

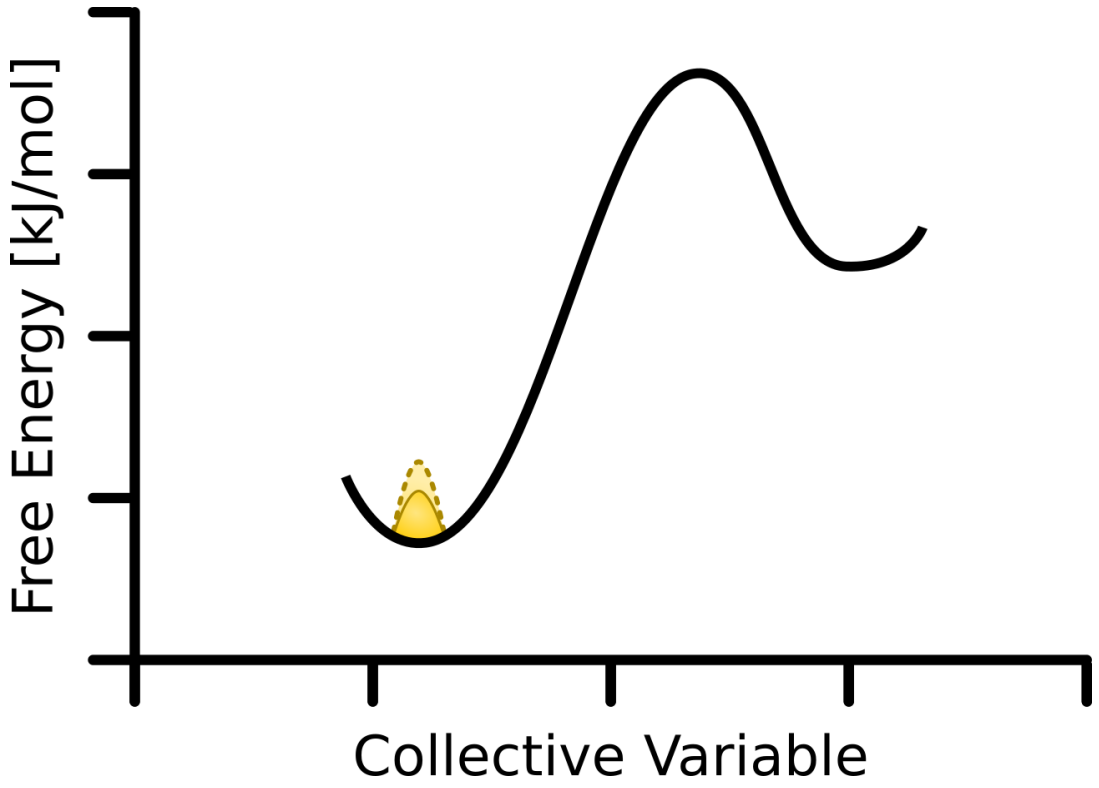
SIGMA: Gaussian width



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Metadynamics

HEIGHT: Gaussian height



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Well-Tempered Metadynamics

```
METAD ...
    LABEL=mtd
    ARG=dist

    SIGMA=0.05
    HEIGHT=1
    PACE=500
    TEMP=310
    BIASFACTOR=5

    GRID_MIN=-0.5
    GRID_MAX=0.5
    GRID_SPACING=0.01

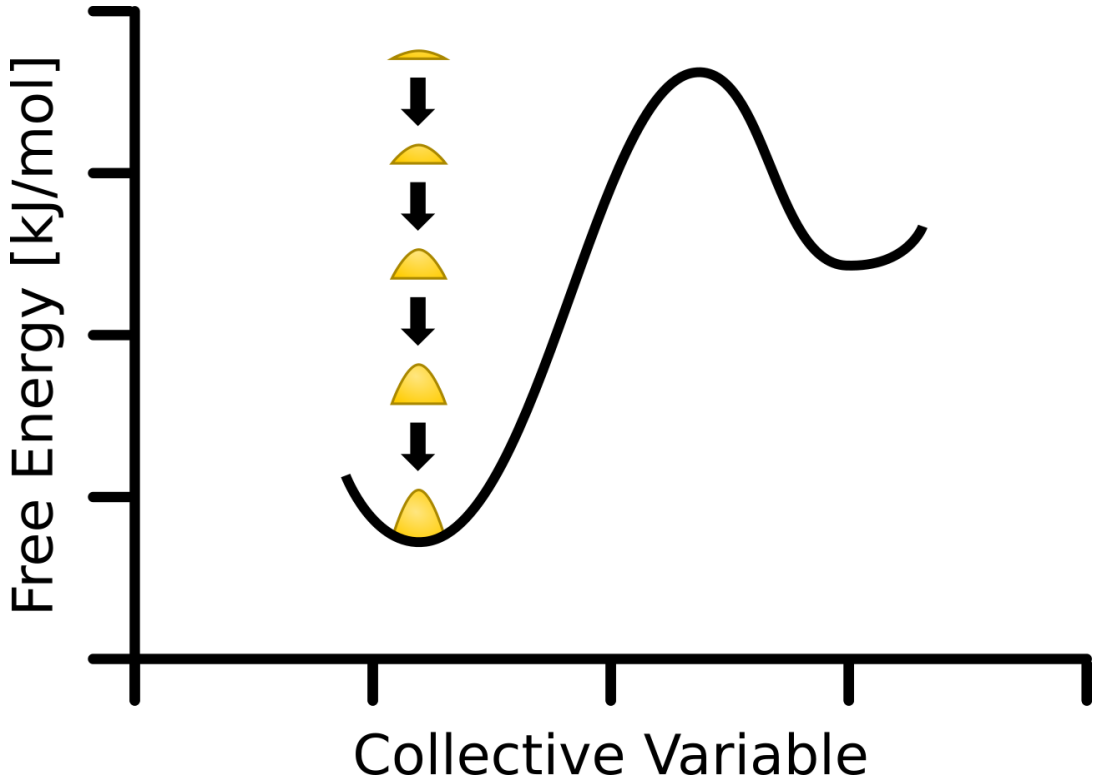
    FILE=HILL
...
```

dat (plumed)

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Well-Tempered Metadynamics

BIASFACTOR: Scales the height of free energy barriers, influencing which regions of the landscape can be explored



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Well-Tempered Metadynamics

Relative distance from the box center along the Z-axis

```
c: CENTER ATOMS=1-3200 NOPBC

d: DISTANCE ATOMS=c,1 SCALED_COMPONENTS
dist: CUSTOM ARG=d.c FUNC=x PERIODIC=-0.5,0.5

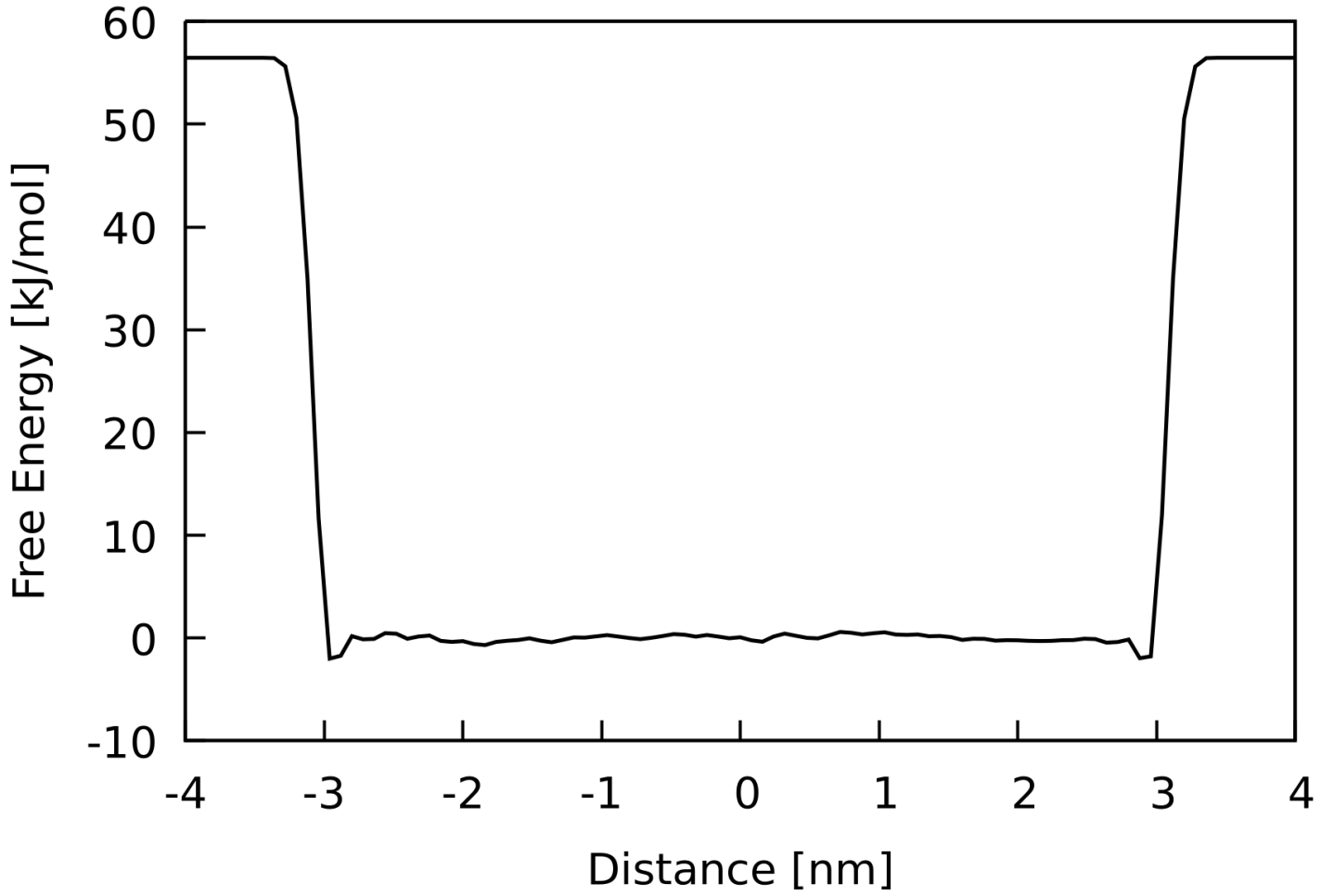
METAD …

lwall: LOWER_WALLS ARG=dist AT=-0.4 KAPPA=1000
uwall: UPPER_WALLS ARG=dist AT=0.4  KAPPA=1000
```

dat (plumed)

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Well-Tempered Metadynamics



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Well-Tempered Metadynamics

Boundary Correction for Metadynamics

Regardless of the metadynamics variant employed, the finite spatial resolution of the Gaussian bias leads to systematic errors near steep walls or mathematical boundaries of CVs. These errors accumulate over the course of the simulation, hindering correct sampling in these regions, which can also cause simulation instabilities during extended runs. As a result, the accuracy of the calculated free energy in these areas can be compromised (94,95) (Figure 5B; left plot). To address this issue, we implemented a boundary correction in Colvars across arbitrary dimensions, based on concepts from previous research. (94,96,97) The approach consists in applying reflecting conditions at the boundaries (94,97) and removing biasing forces components along CVs where exceed their specified boundary limits (96) (i.e., when the tails of the Gaussians extend beyond themathematical limits of the CVs). This boundary condition can be enabled in Colvars with the keyword useHillsReflection. When activated, additional Gaussians are placed outside the boundaries at locations reflected relative to the original ones and with the same height and width of the original Gaussians (Figure 5A). To improve efficiency, this condition is applied only to Gaussians that are within a cutoff distance from a boundary (by default 6 times the Gaussians sigma, which can be customized with the keyword reflectionRange).

https://doi.org/10.1021/acs.jpcb.4c05604

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Well-Tempered Metadynamics

Distance from the box center

```
c: CENTER ATOMS=1-3200 NOPBC

d: DISTANCE ATOMS=c,1

METAD …

uwall: UPPER_WALLS ARG=d AT=3.0 KAPPA=8000
```

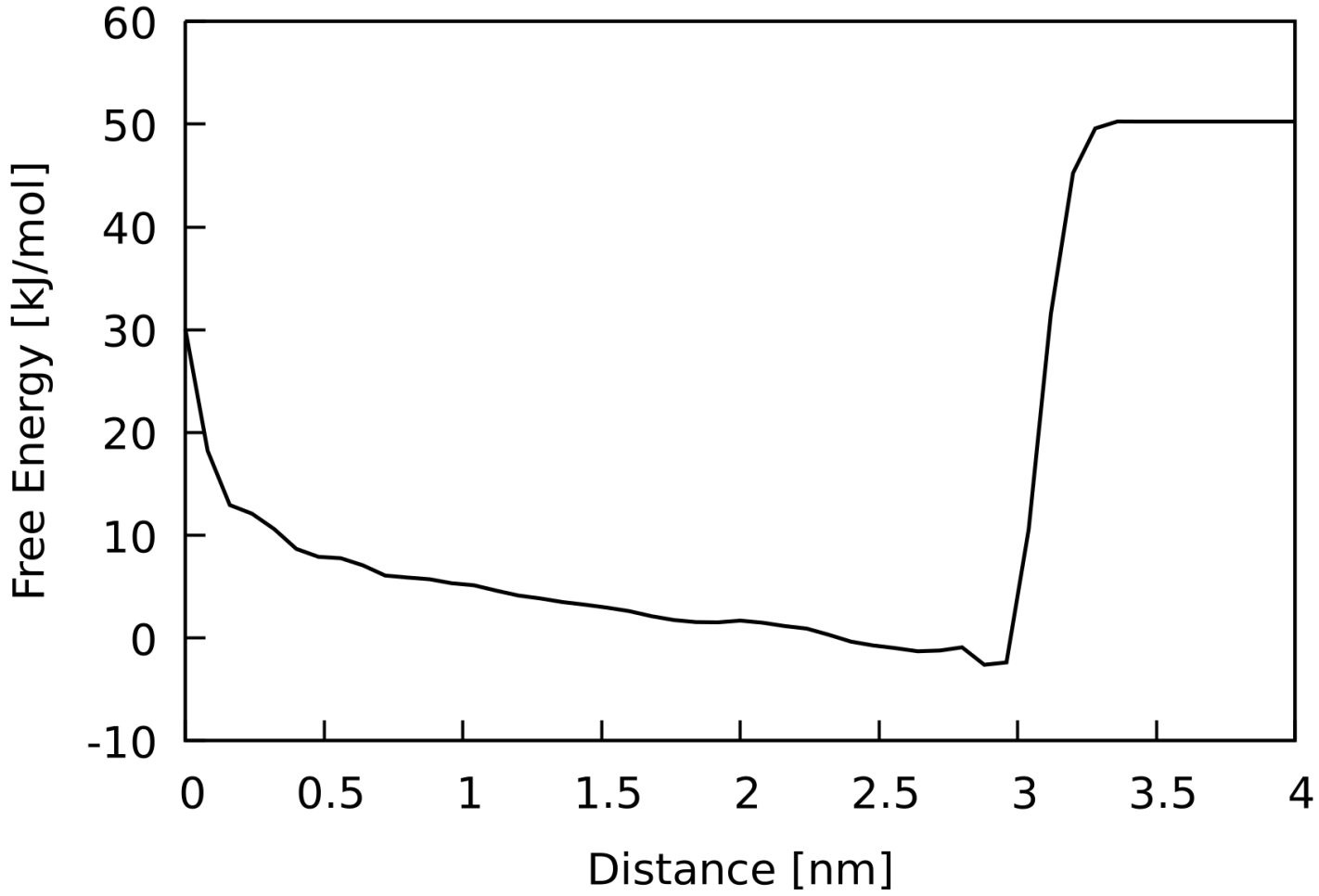
dat (plumed)

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Collective Variable (CV)

- A parameter that describes the progress of a process
  - Reduces the complexity of many-particle systems to a few key degrees of freedom
  - Must accurately represent the process or transition of interest
- CV must be differentiable, smooth, and continuous
- Examples of common collective variables:
  - Distance between atoms/beads or molecules
  - Angle or dihedral angle
  - Coordination number
  - Radius of gyration

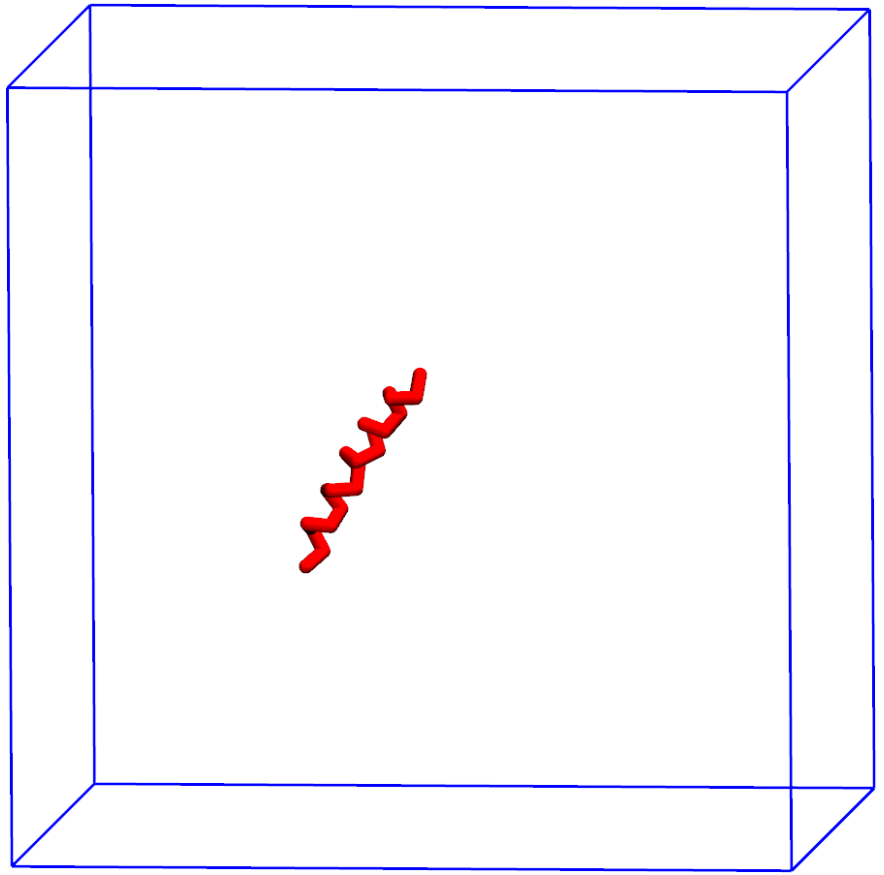
Well-Tempered Metadynamics



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The Toy System

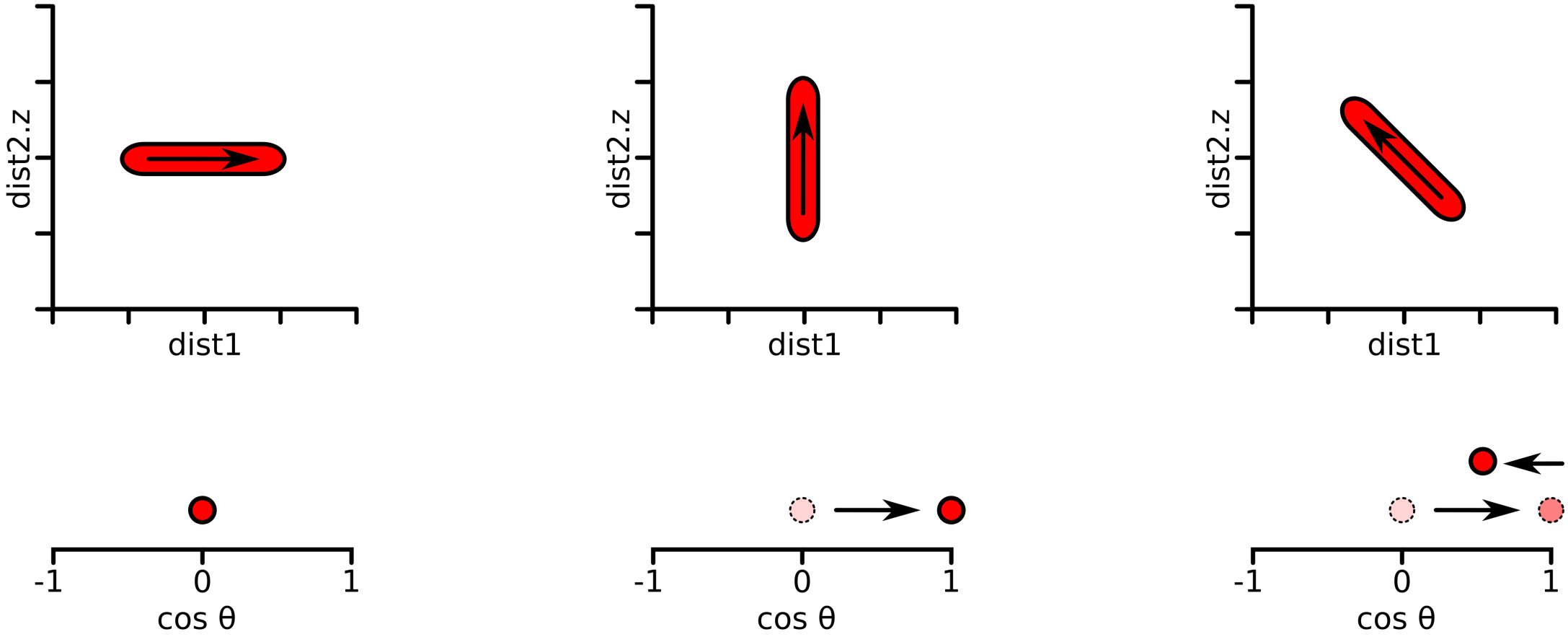
A helical peptide in water



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Calculating the Cosine of an Angle

Collective variable must be continuous



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Well-Tempered Metadynamics

```
dist1: DISTANCE ATOMS=1,21
dist2: DISTANCE ATOMS=1,21 COMPONENTS
cos_angle: CUSTOM ARG=dist1,dist2.z FUNC=y/x PERIODIC=NO

METAD ...
    LABEL=mtd
    ARG=cos_angle

    SIGMA=0.05
    HEIGHT=1
    PACE=500

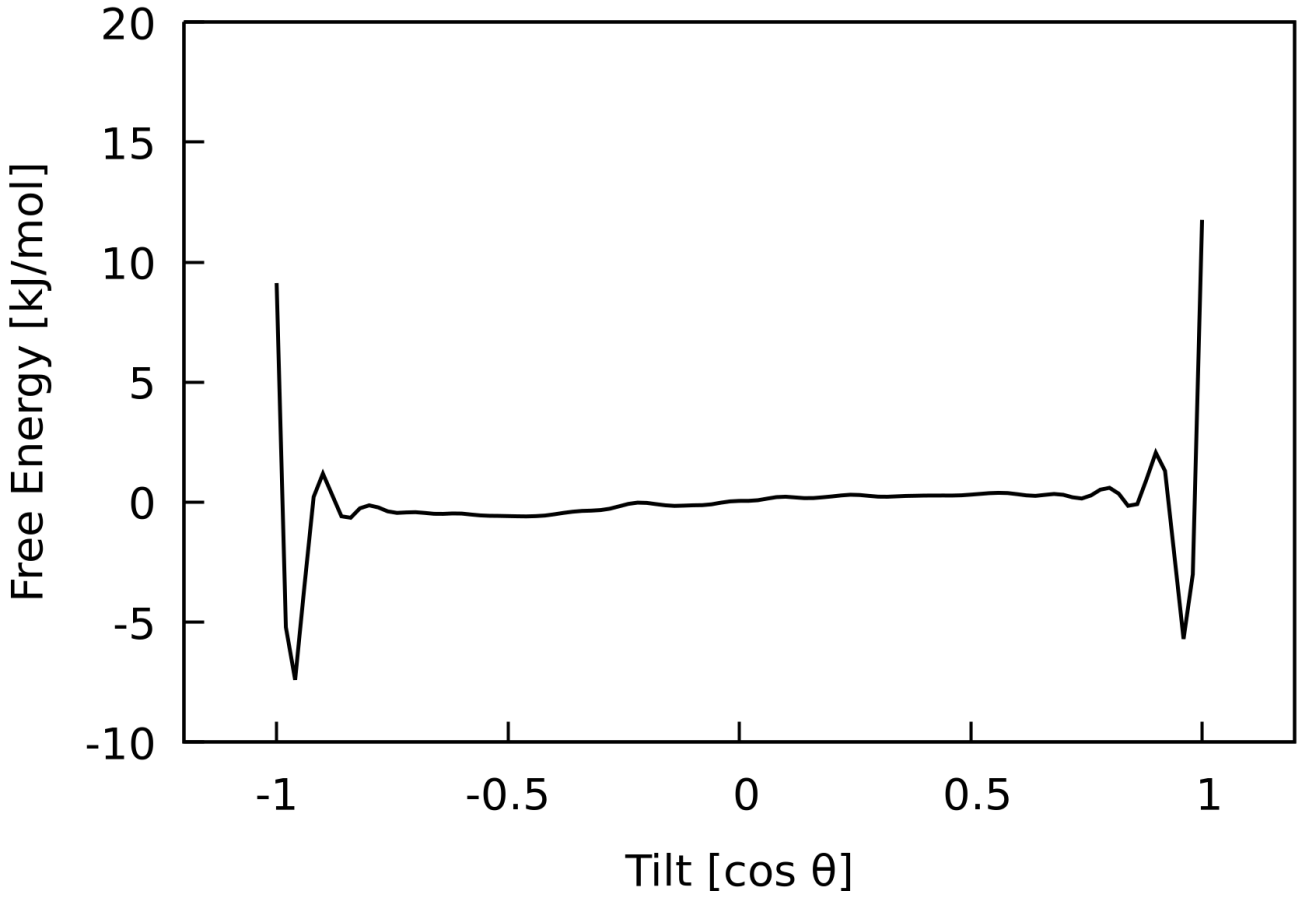
    TEMP=310
    BIASFACTOR=10

[…]
```

dat (plumed)

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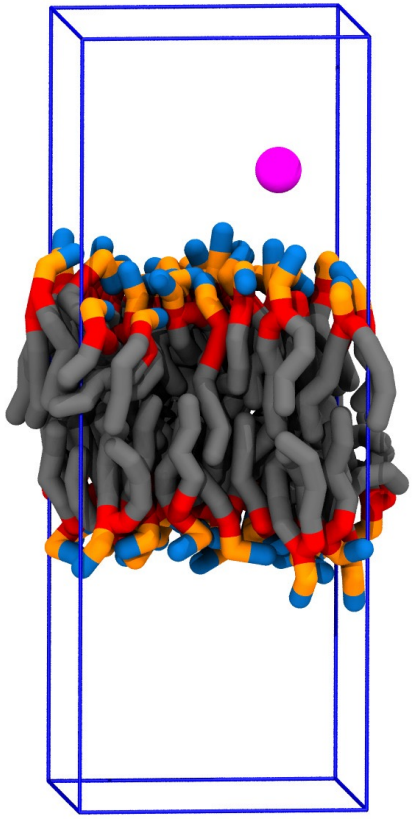
Well-Tempered Metadynamics



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The Toy System

Ion permeation across a lipid bilayer



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Well-Tempered Metadynamics

```
METAD ...
    LABEL=mtd
    ARG=distz

    SIGMA=0.01
    HEIGHT=2.5
    PACE=500
    TEMP=310
    BIASFACTOR=???

    GRID_MIN=-0.5
    GRID_MAX=0.5
    GRID_SPACING=0.002

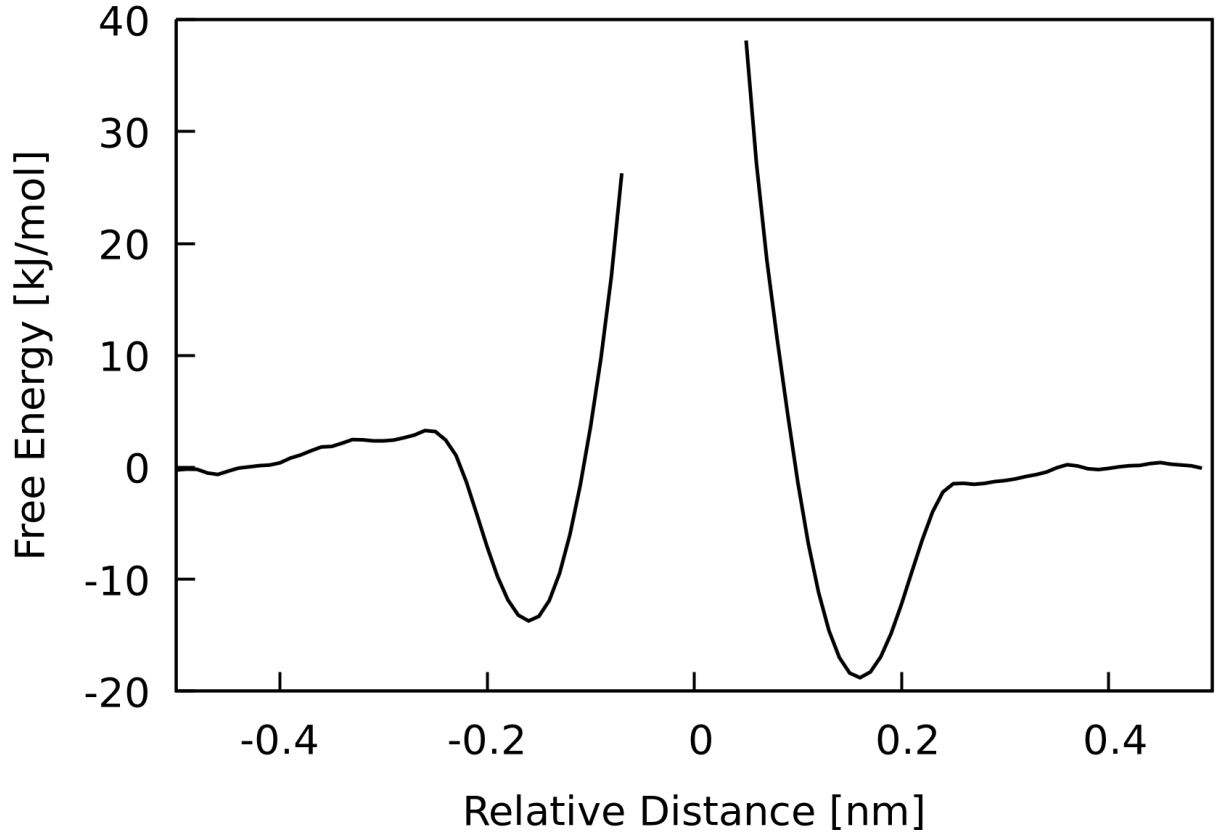
    FILE=HILL
...
```

dat (plumed)

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Free Energy from an Unbiased Simulation

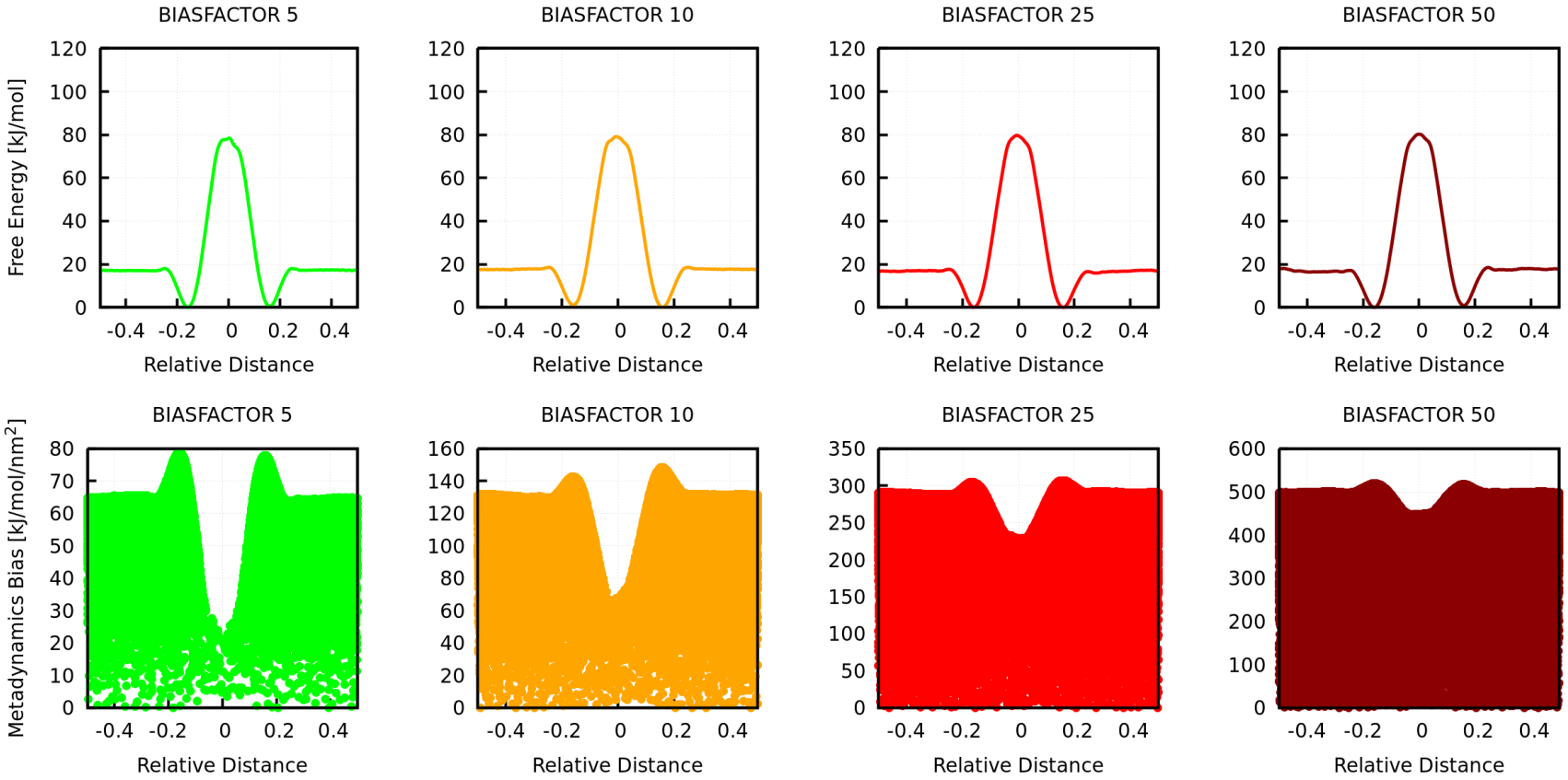
Simulation with a duration of 15 μs



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Free Energy from Metadynamics

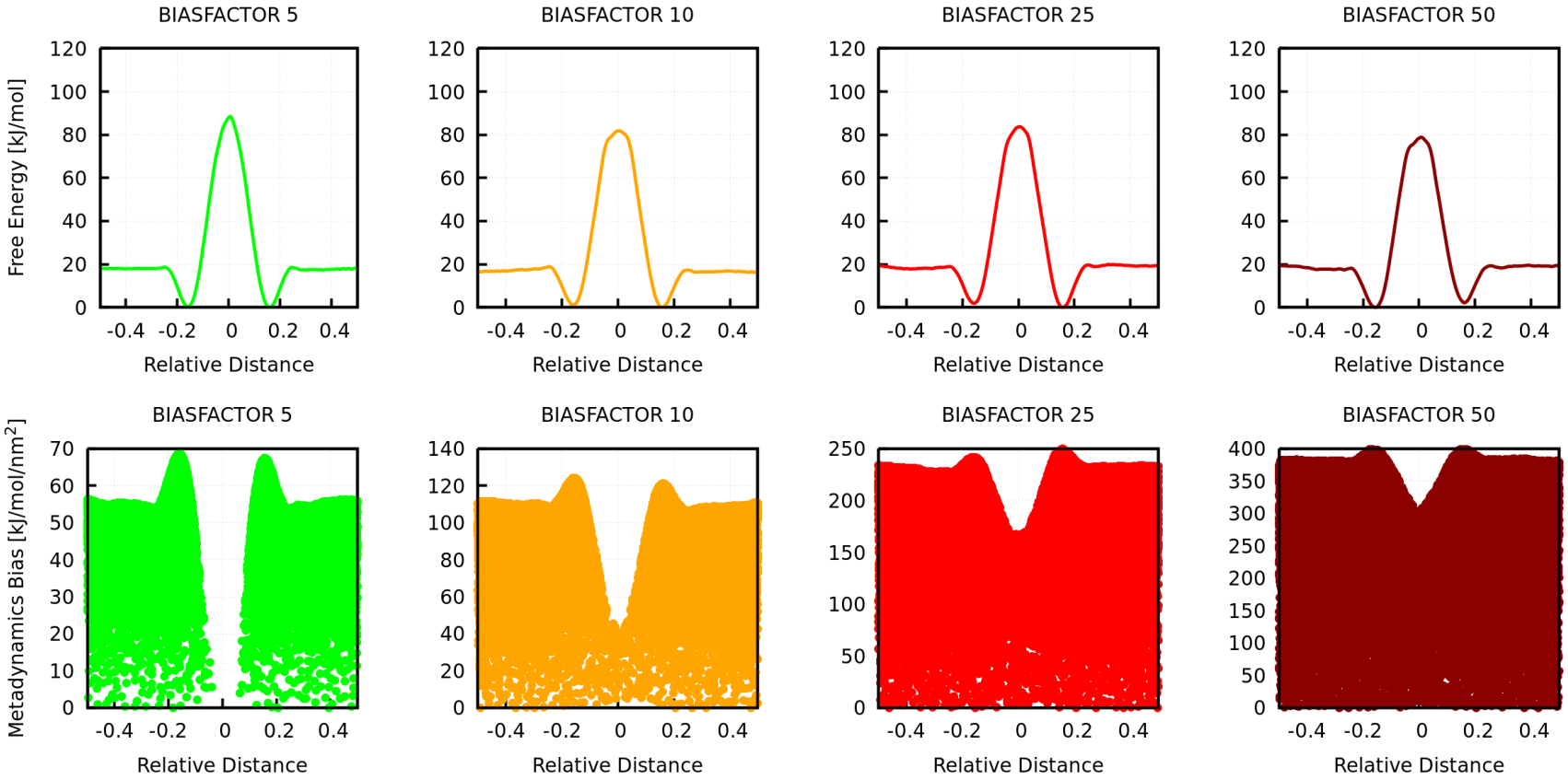
Simulation with a duration of 15 μs



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Free Energy from Metadynamics

Simulation with a duration of 5.5 μs



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Metadynamics with Multiple CVs

… with exponentially increasing computational cost

```
METAD ...
    LABEL=mtd
    ARG=distz,distx

    SIGMA=0.01,0.01
    HEIGHT=2.5
    PACE=500

    TEMP=310
    BIASFACTOR=10

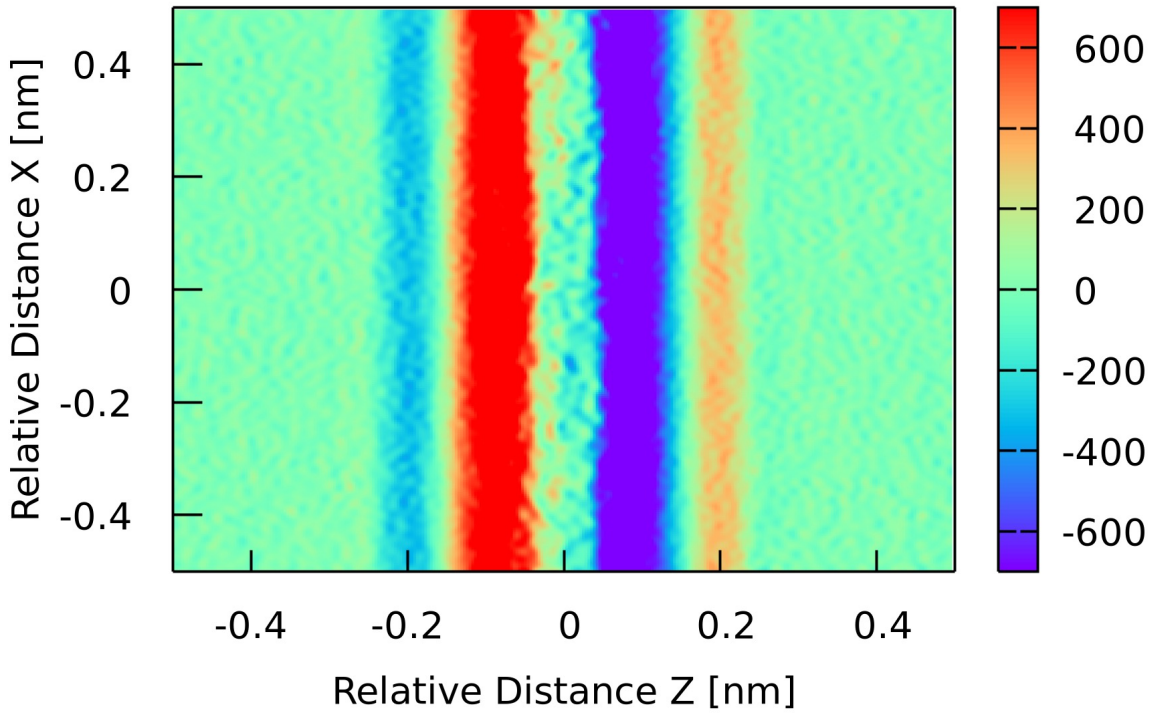
    GRID_MIN=-0.5,-0.5
    GRID_MAX=0.5,0.5
    GRID_SPACING=0.002,0.002
[…]
```

dat (plumed)

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Metadynamics with Multiple CVs

Plotting the results with GNUPLOT




```
set xlabel "Relative Distance Z [nm]"
set ylabel "Relative Distance X [nm]"

set xrange [-0.5:0.5]
set yrange [-0.5:0.5]

set view map

set pm3d interpolate 0,0
set palette rgbformulae 33,13,10

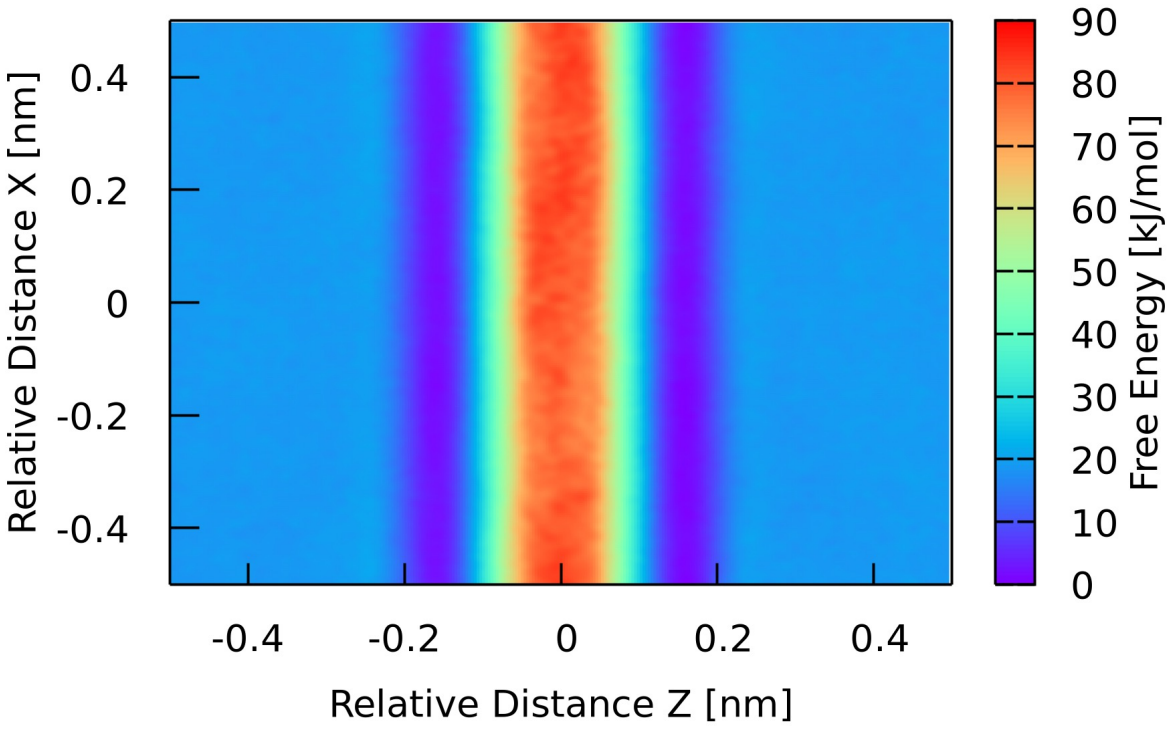
set output "fes_2D.png"
splot 'fes.dat' w pm3d lc palette
```

gnuplot

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Metadynamics with Multiple CVs

Plotting the results **correctly** with GNUPLOT




```
set xlabel "Relative Distance Z [nm]"
set ylabel "Relative Distance X [nm]"

set xrange [-0.5:0.5]
set yrange [-0.5:0.5]
set cbrange [0:90]

set view map

set pm3d interpolate 0,0
set palette rgbformulae 33,13,10
```

gnuplot

```
set output "fes_2D.png"
splot 'fes.dat' u 1:2:3:3 w pm3d […]
```


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Task 1

Using `sum_hills`

- Sum the Gaussians in the file HILL_1D
  - Shift the minimum to zero
  - Set 100 bins for the calculation
- Sum the Gaussians in the file HILL_2D
  - Shift the minimum to zero
  - Set 100 bins for the calculation
  - Integrate out the second collective variable
  - Compare both profiles

Task 1

Setting up a well-tempered metadynamics simulations

- Use XZTORSIONS as the collective variable
- Setup well-tempered metadynamics

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